

# Asymptotically Optimal Mixed-State Cloning

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## Abstract

Unknown quantum states can only be cloned approximately. While the optimal pure-state cloner has long been known, the mixed-state problem has remained open. We determine sharp global asymptotic cloning fidelity for a cloner that receives  $n$  copies of a  $d$ -dimensional full-rank non-degenerate mixed state and produces  $m_n$  copies, with  $m_n/n \rightarrow \gamma > 1$ . For a known spectrum  $p_1 > \dots > p_d > 0$ , the optimal asymptotic cloning fidelity on its unitary orbit is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}}, \quad q_{ij} = \frac{p_j}{p_i}.$$

For unknown spectrum, one spectrum-independent cloner attains

$$F_{\text{univ}}(\gamma, p) = \left( \frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} F_{\text{orb}}(\gamma, p),$$

uniformly on compact subsets of the simple full-rank chamber, and is minimax optimal. The two factors separate the costs of reproducing the unknown spectrum and eigenbasis. Our explicit channels combine classical processing of Schur–Weyl labels with a covariant Cartan-sector channel. Local asymptotic normality and a sharp Gaussian fidelity bound against unrestricted channels give the matching converses. For flat projector states  $\rho = P/r$ , with  $P$  a rank- $r$  orthogonal projector, we separately prove the optimum  $\gamma^{-r(d-r)/2}$  by exact compression and a finite-dimensional converse. Finally, we compute the exact asymptotic fidelity of purify–clone–trace (PCT) at simple full-rank spectra and an upper bound for projector states. At fixed gain, these establish strict superiority of our spectrum-independent cloner at every simple full-rank spectrum and of our projector cloner for  $1 < r < d$ .

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# 1 Introduction

The no-cloning theorem rules out perfect copying of an arbitrary unknown quantum state [WZ82, Die82]. Approximate cloning asks how well a channel can transform  $n$  identical inputs into an output resembling  $m$  ideal copies. We call the ratio  $m/n$  as the cloning *gain*. We measure cloning fidelity globally with the ideal product target  $\rho^{\otimes m}$ , to distinguish from the problem of broadcasting [BCFJS96] where we often judge the marginal fidelity. The goal is to determine the optimal cloning fidelity for any given gain, and to construct a channel that attains it. The pure-state problem was solved in the 1990s, and the optimal cloner is known to be a covariant channel that symmetrically embeds the input into the output space [Wer98, KW99]. The mixed-state problem has remained open [SIGA05], with only partial results for restricted families of states or small gain [BDMP05, BDMP06, DF07, FLS07, Kaz10, LTHC26, FGSS26, JSO26]. We resolve the mix-state cloning problem by determining the optimal cloner under global product fidelity for any simple<sup>1</sup> full-rank spectra at any fixed asymptotic gain.

For mixed states, the eigenvalues and eigenbasis may both be unknown, and these two kinds of information behave differently under repeated sampling. We therefore approach the problem in two steps. In the known-spectrum problem, the eigenvalues are supplied and the state ranges over one unitary orbit; cloning must reproduce only the unknown eigenbasis. In the unknown-spectrum problem, a single spectrum-independent sequence of channels must handle both eigenvalues and eigenbasis, uniformly over compact subsets of the chamber of simple full-rank spectra. This is not merely the first problem preceded by spectrum estimation: the measured spectrum retains fluctuations of order  $n^{-1/2}$ , which must themselves be converted from  $n$  to  $m$  copies.

We solve both problems in the asymptotic limit for a sequence of cloning tasks to produce  $m_n$  clones from  $n$  copies of an unknown state, such that  $m_n/n \rightarrow \gamma > 1$ . For known spectrum, we determine the optimal global fidelity on each unitary orbit. For unknown spectrum, we construct one universal cloner, determine its asymptotic cloning fidelity uniformly on compact subsets of the chamber of simple full-rank spectra, and determine the minimax value over the compact spectral sets (cf. Theorem 1.2). The unknown-spectrum value factors into the known-spectrum optimum and a spectrum-independent classical factor. We also solve cloning of flat rank- $r$  projector states  $\rho_P = P/r$ , with  $P$  any rank- $r$  orthogonal projector. Finally, we analyze the actual output of an alternative cloning protocol [LTHC26, FGSS26, JSO26], purify-clone-trace (PCT), obtaining its exact fidelity at simple full-rank spectra and a strict upper bound for projector states. We show a strict improvement over PCT on both families, beyond a comparison with its lower guarantees in the literature.

Two complementary decompositions drive the solution. Schur–Weyl duality separates the spectral information in the Young label from the eigenbasis information in the conditional irreducible state [KW01]. Our cloners process the label classically and apply a multiplicity-one Cartan-sector channel to the irreducible state [PRV67, FH91]. Local asymptotic normality (LAN) gives the corresponding separation into a classical Gaussian shift and displaced thermal oscillator modes. A cloner exceeding our orbit or compact-family minimax value would induce a channel exceeding the corresponding Gaussian dilation or amplification optimum. Our converse uses the existing LAN framework and a sharp fidelity bound proved here for arbitrary, possibly non-Gaussian and correlated channels. A unified flat-prior argument establishes this bound; the quantum optimum is attained by the product of least-noise amplifiers. Projector states instead admit exact Schur-sector compression and a direct finite-dimensional converse.

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<sup>1</sup>We call a full-rank spectrum *simple* (i.e., non-degenerate) if its eigenvalues are distinct.

## 1.1 Results

Fix an integer  $d \geq 2$  and positive integers  $m_n$  such that

$$\gamma_n := \frac{m_n}{n} \longrightarrow \gamma > 1, \quad r_d := \binom{d}{2}. \quad (1)$$

In particular,  $m_n > n$  for all sufficiently large  $n$ .<sup>2</sup> Let  $\mathbf{A}_d$  be the affine hyperplane  $\mathbf{A}_d := \{x \in \mathbb{R}^d : \sum_i x_i = 1\}$ . Let

$$\Delta_d^\circ := \{p \in \mathbf{A}_d : p_1 > \dots > p_d > 0\}, \quad \rho_p = \text{diag}(p_1, \dots, p_d), \quad \rho_{p,g} = g\rho_p g^*. \quad (2)$$

For positive trace-class operators  $A, C$ , not necessarily normalized, we write

$$F(A, C) = \|\sqrt{A}\sqrt{C}\|_1. \quad (3)$$

For states, this is the root (unsquared) Uhlmann fidelity. For nonnegative functions or measures, the same notation denotes their Hellinger affinity; in particular,  $F(P, Q) = \sum_x \sqrt{P(x)Q(x)}$  on a countable space. The target in every statement below is the entire product state  $\rho_{p,g}^{\otimes m_n}$ . For  $p \in \Delta_d^\circ$ , define

$$q_{ij} = \frac{p_j}{p_i}, \quad f_\gamma(q) = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q}, \quad (4)$$

and

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_\gamma(q_{ij}), \quad F_{\text{cl}}(\gamma, d) = \left( \frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2}, \quad F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p). \quad (5)$$

The factor  $F_{\text{orb}}$  measures the cost of amplifying the unknown eigenbasis. The *classical fidelity factor*  $F_{\text{cl}}$  accounts for the additional cost of reproducing the unknown spectral fluctuations.

**Known spectrum** Fixing  $p$  isolates the genuinely noncommutative part of the problem. The target distribution of Young labels is then known and can be sampled exactly at size  $m_n$ ; only the conditional representation sectors, which encode the eigenbasis, need to be cloned.

**Theorem 1.1** (Known-spectrum optimum). *For every  $p \in \Delta_d^\circ$ ,*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \mathbf{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = F_{\text{orb}}(\gamma, p), \quad (6)$$

where the supremum is over all quantum channels from  $n$  to  $m_n$  qudits. The explicit sequence of channels  $\mathcal{C}_n^p$  defined in [Section 2.2](#) asymptotically attains the displayed value.

Thus every pair  $i < j$  contributes one displaced-thermal amplification factor  $f_\gamma(p_j/p_i)$ . The Schur–Cartan cloner attains their product, while fixed-spectrum LAN shows that any better finite-dimensional cloner would violate the Gaussian amplification optimum proved here.

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<sup>2</sup>At the finitely many exceptional indices with  $m_n \leq n$ , our cloners are defined as the exact partial trace channel (and the identity when  $m_n = n$ ). This convention does not affect any limit, and all construction formulas below concern  $m_n > n$ .

**Unknown spectrum** If  $p$  is not supplied, the target Young-label distribution can no longer be sampled directly. The measured input label determines the spectrum only at the  $n^{-1/2}$  scale, and converting  $n$  copies into  $m_n \sim \gamma n$  copies requires dilating this classical fluctuation. This produces the spectrum-independent classical factor  $F_{\text{cl}}(\gamma, d)$ .

**Theorem 1.2** (Unknown-spectrum cloner fidelity and compact-set optimum). *There exists a sequence of spectrum-independent,  $U(d)$ -covariant channels  $\mathcal{C}_n^{\text{univ}}$ , constructed in [Section 2.3](#), with the following properties.*

(a) For every compact  $K \subseteq \Delta_d^\circ$ ,

$$\lim_{n \rightarrow \infty} \sup_{\substack{p \in K \\ g \in U(d)}} \left| F\left(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) - F_{\text{univ}}(\gamma, p) \right| = 0. \quad (7)$$

(b) Let  $K \subseteq \Delta_d^\circ$  be nonempty and satisfy  $K = \overline{\text{int}_{\mathbf{A}_d} K}^{\mathbf{A}_d}$ . Then

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in U(d)}} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = \inf_{p \in K} F_{\text{univ}}(\gamma, p), \quad (8)$$

where the supremum is over all quantum channels from  $n$  to  $m_n$  qudits.

Part (a) describes the performance of one prescribed cloner on every compact  $K$ , including singletons and lower-dimensional sets; no thickness condition is needed. Part (b), by contrast, identifies the optimum over all channels. In (8), the set  $K$  may inform the channel design, but the actual  $p \in K$  and  $g \in U(d)$  remain unknown. Unitary covariance removes the dependence on  $g$ , and the least favorable spectrum gives the infimum on the right.

The additional condition in part (b) is a sufficient thickness assumption: it says that the interior of  $K$  is dense in  $K$ , with interior taken in the full  $(d-1)$ -dimensional spectral hyperplane  $\mathbf{A}_d$ , not in a possibly smaller affine hull of  $K$ . Around each interior point, the allowed spectra vary in all  $d-1$  independent directions. The converse uses these neighborhoods to obtain the classical factor  $F_{\text{cl}}(\gamma, d)$ . Density of the interior and continuity of  $F_{\text{univ}}(\gamma, p)$  then give the infimum over all of  $K$ ; see [Section 3.4](#).

Part (b) cannot hold for arbitrary compact sets. For instance, if  $K = \{p\}$ , the spectrum is known, and [Theorem 1.1](#) gives the larger optimum  $F_{\text{orb}}(\gamma, p) > F_{\text{univ}}(\gamma, p)$ . Part (a) still describes our spectrum-independent cloner correctly on this singleton, but that cloner does not exploit the known spectrum and is not optimal there. Other lower-dimensional spectral families may likewise have different classical factors.

A natural admissible example of  $K$  is one that keeps the smallest eigenvalue and every adjacent spectral gap uniformly positive. For  $0 < \kappa < 2/[d(d+1)]$ , set  $K$  to be

$$\Delta_{d,\kappa} = \{p \in \mathbf{A}_d : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa \ (i < d)\}.$$

This is a nonempty compact subset of  $\Delta_d^\circ$  and satisfies  $\Delta_{d,\kappa} = \overline{\text{int}_{\mathbf{A}_d} \Delta_{d,\kappa}}^{\mathbf{A}_d}$ , so (8) applies.<sup>3</sup>

The common Schur–Cartan architecture is summarized in [Fig. 1](#). After the Schur transform, the Young label  $\mu$  carries the spectral information, while the conditional irrep state carries the unknown eigenbasis. A classical kernel chooses an output label  $\nu$ , and the common covariant sector channel maps the conditional state into the  $\nu$ -sector before the  $m_n$ -system output is reconstructed. For

<sup>3</sup>To verify nonempty relative interior, choose  $\kappa' \in (\kappa, 2/[d(d+1)])$  and set  $p_i = 1/d + ((d+1)/2 - i)\kappa'$ . At the endpoint  $\kappa = 2/[d(d+1)]$ , the constraints force a singleton, so the compact-set theorem no longer applies.

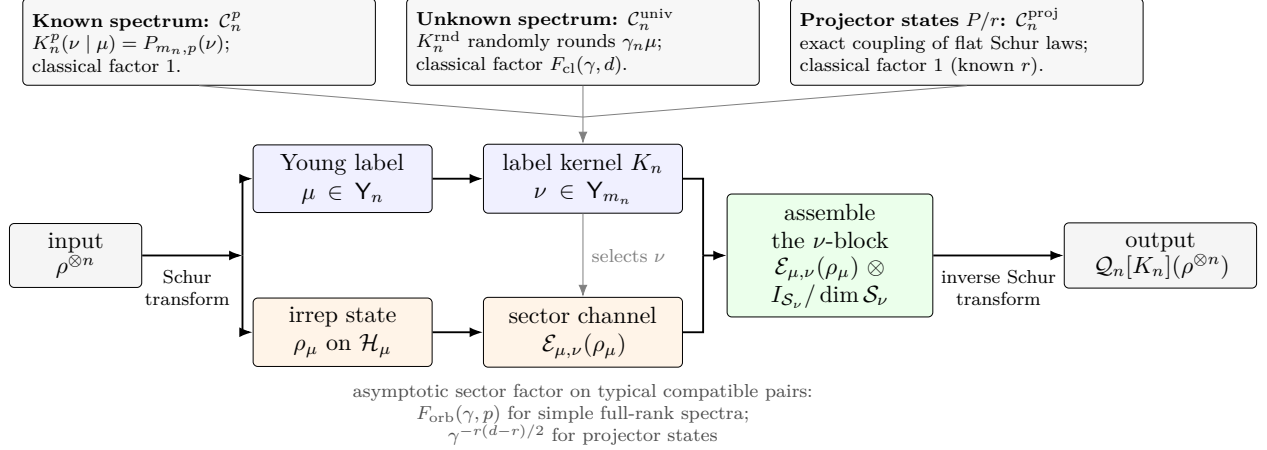


Figure 1: Common construction of  $\mathcal{C}_n^p$ ,  $\mathcal{C}_n^{\text{univ}}$ , and  $\mathcal{C}_n^{\text{proj}}$ . The input is  $\rho = \rho_{p,g}$  for the simple full-rank families or  $\rho = P/r$  for projector states. The Schur transform measures the Young label  $\mu$  and retains the normalized conditional irrep state  $\rho_\mu$ . The three choices of label kernel are shown above; the projector kernel couples the exact input and target flat Schur laws. All three cloners apply the same sector channel  $\mathcal{E}_{\mu,\nu}$  and restore a maximally mixed Specht factor. The displayed classical and sector factors multiply to the attained asymptotic fidelities  $F_{\text{orb}}$ ,  $F_{\text{univ}}$ , and  $\gamma^{-r(d-r)/2}$  in Theorems 1.1 to 1.3, respectively.

known  $p$ , the label kernel samples the exact target law  $P_{m_n,p}$  independently of  $\mu$ ; the spectrum-independent kernel instead randomly rounds  $\gamma_n \mu$ . Their classical factors are therefore 1 and  $F_{\text{cl}}$ , while the common sector factor is  $F_{\text{orb}}$ , giving the exact asymptotic fidelities  $F_{\text{orb}}$  and  $F_{\text{univ}}$  for the known- and unknown-spectrum cloners, respectively. The details are developed in Section 2.

We did not attempt to solve the general rank-deficient problem, but we can treat the special case of flat projector states.

**Theorem 1.3** (Cloning projector states). *For fixed  $1 \leq r < d$ , let  $\text{Gr}(r, d)$  denote the set of rank- $r$  orthogonal projectors on  $\mathbb{C}^d$ . Then*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{P \in \text{Gr}(r, d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) = \gamma^{-r(d-r)/2}. \quad (9)$$

The limit is attained by our cloner  $\mathcal{C}_n^{\text{proj}}$  constructed in Section 4.2. For  $r = 1$ , the value  $\gamma^{-(d-1)/2}$  recovers the optimal asymptotic global fidelity for pure-state cloning [Wer98]. For  $r = d$ , the family contains only  $I_d/d$ , and the value is one.

The cloner  $\mathcal{C}_n^{\text{proj}}$  uses the same Schur–Cartan architecture as  $\mathcal{C}_n^p$  and  $\mathcal{C}_n^{\text{univ}}$ , with a different transition rule between Young diagrams, also shown in Fig. 1. For projector states the classical factor is 1, while the asymptotic sector factor is  $\gamma^{-r(d-r)/2}$ . Both achievability and the converse are proved separately in Section 4.

## 1.2 Remarks

**Dependence on spectral mixedness** The product formula makes precise one sense in which a more mixed state is easier to clone. For fixed  $\gamma > 1$ ,

$$f'_\gamma(q) = \frac{(\gamma - 1)^2}{2\sqrt{q(\gamma - 1 + q)}(\sqrt{\gamma(\gamma - 1 + q)} + \sqrt{q})^2} > 0, \quad 0 < q < 1.$$

Consequently, if  $p, p' \in \Delta_d^\circ$  satisfy

$$\frac{p'_j}{p'_i} \geq \frac{p_j}{p_i} \quad \text{for every } i < j, \quad (10)$$

that is, if every logarithmic spectral gap contracts in passing from  $p$  to  $p'$ , then

$$F_{\text{orb}}(\gamma, p') \geq F_{\text{orb}}(\gamma, p), \quad F_{\text{univ}}(\gamma, p') \geq F_{\text{univ}}(\gamma, p), \quad (11)$$

with strict inequality if any ratio is strictly larger.<sup>4</sup>

**Comparison with PCT** The quantitative comparison with the purify–clone–trace protocol is developed in [Section 5](#). At every fixed  $\gamma > 1$ , our spectrum-independent cloner has strictly larger asymptotic fidelity than PCT at every simple full-rank spectrum, and our projector-state cloner is strictly better than PCT for  $1 < r < d$ . [Figure 2](#) previews both separations; [Section 5](#) and [Appendix A](#) give the exact formulas, their sample complexities, and the order-of-limits caveat.

**Limitations** Apart from flat projector states, our theorems require a simple full-rank spectrum. We do not treat nonflat rank-deficient spectra, positive degeneracies, or neighborhoods in which the rank changes. Our asymptotic estimates are established at fixed gain  $\gamma > 1$ ; we do not control the joint regime in which the sample size grows and  $\gamma \downarrow 1$ . We also do not determine the exact finite-copy optimum. Expected extensions to degenerate and rank-deficient spectra, and the noncommuting limits at degeneracies, are discussed in [Section 5.3](#).

### 1.3 Earlier and related work

**Pure-state cloning.** Early universal qubit cloners [[BH96](#), [GM97](#), [BDE+98](#)] were followed by Werner’s solution of global universal cloning in arbitrary dimension [[Wer98](#)] and Keyl–Werner’s optimality result for individual-clone tests [[KW99](#)]. State-dependent global cloning was developed in [[BDE+98](#), [CB99](#)]. For individual clones, the many-output limit is closely tied to estimation [[MP95](#), [BEM98](#), [BA06](#), [CD06](#)]. For the joint output, the relation is subtler: global asymptotic cloning has been analyzed for broad pure-state families, and an optimal measure-and-prepare output need not consist of independent copies of the estimated state [[CY14](#)]. The criterion also matters for coherent-state and phase-covariant cloning [[CKNWW05](#), [KC22](#)]: in particular, Cerf et al. prove Gaussian optimality for joint fidelity, while non-Gaussian operations can improve single-clone fidelity. Reviews include [[SIGA05](#), [Chi11](#), [FWJ+14](#)].

**Mixed-state cloning, broadcasting, and PCT.** The no-broadcasting theorem excludes exact cloning of a noncommuting family of density matrices [[BCFJS96](#), [KH08](#)]. Approximate mixed-state results include global-fidelity, relative-error, and state-dependent bounds for prescribed finite ensembles [[Ras03a](#), [Ras03b](#), [Ras03c](#)]; cloners for symmetric-subspace inputs and identical mixed qubits under one-site shrinking-factor criteria [[Fan03](#), [FLS07](#)]; a restricted mixed-qubit construction under averaged Hilbert–Schmidt loss [[Kaz10](#)]; and entropic limitations on simultaneous approximate cloning and broadcasting [[LW17](#)]. Cirac–Ekert–Macchiavello solved optimal mixed-qubit purification and a cloning/estimation problem under local fidelity [[CEM99](#)]. Multi-input broadcasting and superbroadcasting likewise optimize marginals and permit correlations [[DMP05](#), [BDMP05](#), [BDMP06](#), [DF07](#)]; these criteria do not determine fidelity with the joint product target.

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<sup>4</sup>Condition (10) implies that  $p'$  is majorized by  $p$  but is stronger than majorization: for  $d \geq 3$ , majorization alone need not increase the cloning fidelity.



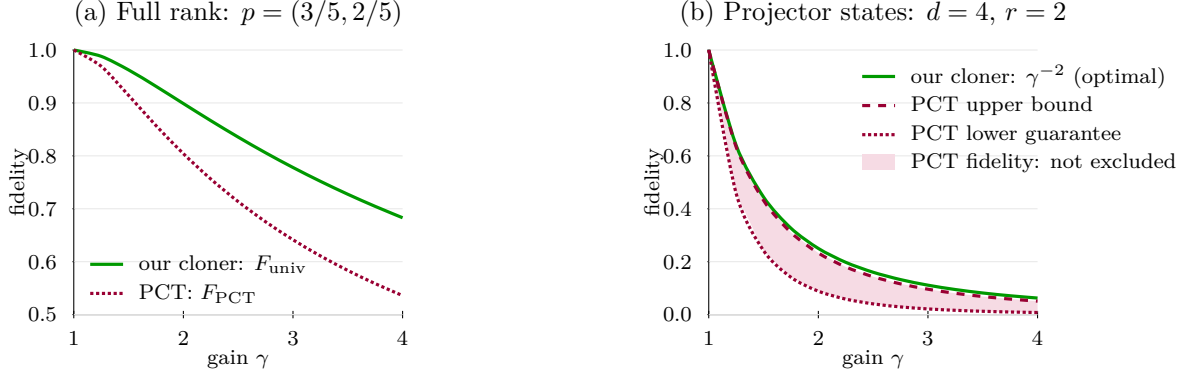


Figure 2: Asymptotic fidelities and bounds as functions of the gain  $\gamma = \lim m_n/n$ . (a) For the simple qubit spectrum  $p = (3/5, 2/5)$ , with neither protocol supplied the spectrum:  $F_{\text{univ}}$  is the exact asymptotic fidelity of our spectrum-independent cloner (Theorem 1.2) and  $F_{\text{PCT}}$  is the exact asymptotic fidelity of PCT (75) (Theorem A.2). (b) For projector states with  $d = 4$  and  $r = 2$ , with both protocols supplied  $r$  but not  $P$ : the solid curve is the optimal value  $\gamma^{-2}$ , attained by our projector-state cloner  $\mathcal{C}_n^{\text{proj}}$  (Theorem 1.3); the dashed curve is the upper bound  $\gamma^{-2}(\sqrt{2\gamma-1}/\gamma)^{1/2}$  on PCT from Proposition A.3, and the dotted curve is the published PCT lower guarantee  $\gamma^{-7/2}$ . The shaded region is the interval not excluded by these two bounds; it is not a confidence band and does not assert that the PCT fidelity converges. Since the dashed curve lies strictly below the solid one for every  $\gamma > 1$ , our projector cloner is strictly better.

Random purification [TWZ25] followed by pure-state cloning gives a global-fidelity guarantee for bounded-rank mixed states [LTHC26]. Concurrent work shows that PCT attains the optimal sample complexity up to constants for squared fidelity  $1 - \epsilon$ , uniformly over states of rank at most  $r$  [FGSS26, JSO26]. These worst-case high-fidelity results complement our exact fixed-gain cloning fidelity results; Section 5 compares them and distinguishes the orders of limits. We show that PCT is strictly suboptimal for every fixed gain  $\gamma > 1$ .

**Gaussian cloning and LAN.** Guță–Matsumoto solved cloning of displaced thermal oscillator states with known temperature under joint norm-distance loss, reducing it to amplification and proving Gaussian optimality by stochastic ordering [GM06]. Bowles–Guță–Adesso used LAN for asymptotically optimal mixed-qubit purification and dilution under global trace-norm loss [BGA11]; related methods solve qudit channel reversal [BGA12]. These norm-distance optima do not by themselves determine our sharp fidelity constants in arbitrary fixed dimension, the separate spectral and orbital costs, or the explicit finite-dimensional Schur–Cartan cloners.

The relevant LAN framework was developed in [GJ07, GK06, KG09, YFG13]. Our fidelity amplification bound is attained by the least-noise amplifier [Cav82, GM06]. Low-rank LAN has a different mode structure [LN24]; we treat the boundary directly only for flat projector states.

## 1.4 Organization

Section 2 constructs the Schur–Cartan cloners and proves achievability from uniform label and sector estimates, established in Appendices B and C. Section 3 combines two-way LAN with the unrestricted Gaussian fidelity converse of Appendix D to prove full-rank optimality. Independently, Section 4 proves the projector-state theorem by exact compression and a finite-dimensional converse.



Section 5 compares these results with PCT and its high-fidelity sample complexity, then discusses remaining open problems. The detailed analysis of PCT is in Appendix A.

## Glossary of notation

We use  $U(d)$  for the unitary group and  $\mathcal{T}_1(\mathcal{H})$  for trace-class operators. The dimension  $d$  is fixed, we denote the gain and asymptotic gain by  $\gamma_n = m_n/n \rightarrow \gamma > 1$ ,  $N$  denotes a generic sample size, and  $r_d = \binom{d}{2}$ . Fidelity uses the unsquared convention,  $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ . Unless stated otherwise, limits are as  $n \rightarrow \infty$  with  $d, \gamma$  fixed;  $o(1)$  is uniform only over the parameters specified in its statement.

| Symbols                                                                                    | Meaning                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A_d, \Delta_d^\circ; K \in \Delta_d^\circ$                                                | The affine hyperplane $\sum_i p_i = 1$ , its strictly ordered positive chamber, and a compact subset of that chamber. Relative interiors and closures are in $A_d$ .                                         |
| $\rho_p, \rho_{p,g}; P \in \text{Gr}(r, d)$                                                | $\rho_p = \text{diag}(p)$ , $\rho_{p,g} = g\rho_p g^*$ ; a rank- $r$ orthogonal projector, giving the state $P/r$ , $1 \leq r < d$ .                                                                         |
| $q_{ij}, f_\gamma(q_{ij})$                                                                 | $q_{ij} = p_j/p_i$ and its one-mode fidelity factor (4).                                                                                                                                                     |
| $F_{\text{orb}}, F_{\text{cl}}, F_{\text{univ}}$                                           | Orbit optimum, classical fidelity factor, and their product $F_{\text{univ}} = F_{\text{cl}}F_{\text{orb}}$ , attained uniformly on compact spectral sets; see (5) and the minimax statement in Theorem 1.2. |
| $Y_N; \mathcal{H}_\lambda, \mathcal{S}_\lambda$                                            | Partitions $\lambda \vdash N$ with at most $d$ rows; the associated unitary and symmetric-group representation spaces.                                                                                       |
| $P_{N,p}, \rho_{p,\lambda}; \mathbf{T}_N(p)$                                               | Young-label law of $\rho_p^{\otimes N}$ , conditional irreducible state, and typical-label set.                                                                                                              |
| $K_n(\nu \mid \mu); \mathcal{Q}_n[K_n]$                                                    | A Markov kernel on Young labels and its covariant channel lift. The spectral set $K$ is distinct from the kernel $K_n$ .                                                                                     |
| $\mathcal{C}_{\mu,\nu}, \mathcal{E}_{\mu,\nu}$                                             | Cartan-sector channel for compatible labels and its extension to all label pairs by a fallback channel.                                                                                                      |
| $\mathcal{M}_n; \mathcal{C}_n^p, \mathcal{C}_n^{\text{univ}}, \mathcal{C}_n^{\text{proj}}$ | A general cloner; our known-spectrum, unknown-spectrum, and projector-state cloners.                                                                                                                         |
| $\mathbf{H}; h, z$                                                                         | Spectral tangent hyperplane $\sum_i h_i = 0$ ; local spectral and orbital coordinates $h \in \mathbf{H}$ , $z \in \mathbb{C}^{r_d}$ .                                                                        |
| $Y_p(z), g_{n,z}^{(p)}; \Omega_L$                                                          | Skew-Hermitian generator at the fixed base spectrum $p$ , $g_{n,z}^{(p)} = e^{Y_p(z)/\sqrt{n}}$ ; bounded local parameter box.                                                                               |
| $\mathcal{F}, \mathcal{F}_{\text{orb}}; \hat{N}$                                           | One-mode Fock space, $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$ , and number operator; distinct from fidelity $F$ and sample size $N$ .                                                          |
| $\tau_p, \tau_p^{(\gamma)}$                                                                | Thermal product in the input and ideal target, and noisy amplifier output at zero displacement.                                                                                                              |
| $\Phi_z^{(p)}, \Psi_z^{(p,\gamma)}; \Theta^{\text{in}}, \Theta^{\text{tar}}$               | Displaced input and ideal target built from $\tau_p$ ; their counterparts including the classical Gaussian component.                                                                                        |

Symbols confined to one proof are defined there.

## 2 Cloner construction and fidelity analysis

The known- and unknown-spectrum protocols use the same finite-dimensional architecture. We first recall the Schur–Weyl decomposition and lift an arbitrary Markov kernel on Young labels to a covariant quantum channel. We then choose the exact target-law kernel for known spectrum and the randomized dilation kernel for unknown spectrum, and combine their label estimates with the common sector analysis. The fidelities of the constructed cloners are evaluated in [Section 2.5](#). We now develop the components summarized in [Fig. 1](#). We start by reviewing some standard facts.

Let  $\mathbf{Y}_N = \{\lambda \vdash N : \ell(\lambda) \leq d\}$ , Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\lambda \in \mathbf{Y}_N} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda, \quad (12)$$

where  $\mathcal{H}_\lambda$  is the irreducible  $U(d)$ -module of highest weight  $\lambda$  and  $\mathcal{S}_\lambda$  is the corresponding Specht module. For  $\rho_p = \text{diag}(p_1, \dots, p_d)$ ,

$$\rho_p^{\otimes N} = \bigoplus_{\lambda \in \mathbf{Y}_N} P_{N,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (13)$$

Here  $\pi_\lambda$  is the irreducible polynomial representation of highest weight  $\lambda$ , acting on  $\mathcal{H}_\lambda$ , and  $s_\lambda(p) = \text{Tr} \pi_\lambda(\rho_p)$  is the corresponding Schur polynomial. For  $\rho_{p,g} = g\rho_p g^*$ , replace  $\rho_{p,\lambda}$  by  $\rho_{p,g,\lambda} = U_\lambda(g)\rho_{p,\lambda}U_\lambda(g)^*$ , where  $U_\lambda(g) = \pi_\lambda(g)$ . The Young law  $P_{N,p}$  depends on  $p$ , but not on  $g$ .

Define the typical Young-label set

$$\mathbf{T}_N(p) = \left\{ \lambda \in \mathbf{Y}_N : \left\| \frac{\lambda}{N} - p \right\|_\infty \leq N^{-1/3} \right\}. \quad (14)$$

**Lemma 2.1** (Uniform Young concentration). *There is a constant  $c_d > 0$ , depending only on  $d$ , such that*

$$\sup_{\substack{p_1 \geq \dots \geq p_d > 0 \\ \sum_i p_i = 1}} P_{N,p}(\mathbf{T}_N(p)^c) \leq e^{-c_d N^{1/3}} \quad (15)$$

for all sufficiently large  $N$ . Equal positive eigenvalues are allowed.

*Proof.* Write  $q = \lambda/N$ . The Weyl dimension formula gives  $\dim \mathcal{H}_\lambda \leq (N+1)^{d(d-1)/2}$ . Every weight  $\xi$  of  $\mathcal{H}_\lambda$  has  $\lambda - \xi$  equal to a nonnegative integral combination of positive roots. Since  $p_j/p_i \leq 1$  for  $i < j$ , this gives  $p^\xi \leq p^\lambda$ . Hence  $s_\lambda(p) \leq \dim \mathcal{H}_\lambda p^\lambda$ . The standard bounds  $\dim \mathcal{S}_\lambda \leq \binom{N}{\lambda_1, \dots, \lambda_d}$  and  $\binom{N}{\lambda_1, \dots, \lambda_d} p^\lambda \leq \exp[-ND(q\|p)]$  give

$$P_{N,p}(\lambda) \leq (N+1)^{d(d-1)/2} \exp[-ND(q\|p)], \quad (16)$$

which is the estimate underlying the Keyl–Werner spectrum large-deviation theorem [\[KW01\]](#).

There are at most  $(N+1)^{d-1}$  partitions of  $N$  with at most  $d$  rows. If  $\lambda \notin \mathbf{T}_N(p)$ , then  $\|q - p\|_1 > N^{-1/3}$ , and Pinsker’s inequality gives  $D(q\|p) \geq \frac{1}{2}N^{-2/3}$ . Therefore

$$P_{N,p}(\mathbf{T}_N(p)^c) \leq (N+1)^{d(d-1)/2+d-1} e^{-N^{1/3}/2}.$$

For all sufficiently large  $N$ , the polynomial factor is at most  $e^{N^{1/3}/4}$ . Hence the asserted bound holds with  $c_d = 1/4$ .  $\square$

## 2.1 Cartan-sector channel and lifting of a label kernel

Let  $\lambda, \theta$  be dominant weights, with unit highest-weight vectors  $v_\lambda, v_\theta$ . The Cartan component  $\mathcal{H}_{\lambda+\theta}$  occurs with multiplicity one in  $\mathcal{H}_\lambda \otimes \mathcal{H}_\theta$ . Let

$$V_{\lambda,\theta} : \mathcal{H}_{\lambda+\theta} \longrightarrow \mathcal{H}_\lambda \otimes \mathcal{H}_\theta, \quad V_{\lambda,\theta} v_{\lambda+\theta} = v_\lambda \otimes v_\theta, \quad (17)$$

be the normalized intertwining isometry. For  $\mu \in \mathbf{Y}_n$  and  $\nu \in \mathbf{Y}_m$ , write  $\nu - \mu \vdash m - n$  when the coordinatewise difference  $\nu - \mu$  is a partition of  $m - n$ , and call such a pair  $(\mu, \nu)$  *compatible*. If this holds, put  $\omega = \nu - \mu$  and define

$$\mathcal{C}_{\mu,\nu}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^* (X \otimes I_{\mathcal{H}_\omega}) V_{\mu,\omega}. \quad (18)$$

The partial trace  $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*)$  commutes with the irreducible  $\text{U}(d)$ -action on  $\mathcal{H}_\mu$ ; hence Schur's lemma and a trace comparison give  $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*) = \frac{\dim \mathcal{H}_\nu}{\dim \mathcal{H}_\mu} I_{\mathcal{H}_\mu}$ . Formula (18) is manifestly completely positive; the identity above makes it trace preserving, while equivariance of  $V_{\mu,\omega}$  gives  $\text{U}(d)$ -covariance.

When  $\mu = (n, 0, \dots, 0)$  and  $\nu = (m, 0, \dots, 0)$ , the three irreps in (18) are symmetric, and the formula is Werner's universal pure-state  $n$ -to- $m$  cloner [Wer98]. Thus the Cartan-sector channel extends Werner's construction from symmetric tensors to arbitrary compatible Schur sectors.

For every pair  $(\mu, \nu) \in \mathbf{Y}_n \times \mathbf{Y}_m$ , define a covariant channel  $\mathcal{E}_{\mu,\nu} : \mathcal{T}_1(\mathcal{H}_\mu) \rightarrow \mathcal{T}_1(\mathcal{H}_\nu)$  by

$$\mathcal{E}_{\mu,\nu}(X) = \begin{cases} \mathcal{C}_{\mu,\nu}(X), & \nu - \mu \vdash m - n, \\ \text{Tr}(X) I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu, & \text{otherwise.} \end{cases} \quad (19)$$

Let  $K_n : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$ , with transition probabilities  $K_n(\nu \mid \mu)$ , be any Markov kernel. Its Schur–Cartan lifting is the following channel, denoted by  $\mathcal{Q}_n[K_n]$ : apply the Schur transform, measure  $\mu$ , discard the input Specht factor, sample  $\nu$  from  $K_n(\cdot \mid \mu)$ , apply  $\mathcal{E}_{\mu,\nu}$ , adjoin  $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$ , and apply the inverse Schur transform. Each step is completely positive and trace preserving, so their composition is a channel. Because the label measurement and sampling are invariant and every map  $\mathcal{E}_{\mu,\nu}$  is covariant,  $\mathcal{Q}_n[K_n]$  is  $\text{U}(d)$ -covariant.

**Proposition 2.2** (Fidelity of a lifted label kernel). *Fix  $p \in \Delta_d^\circ$  and  $g \in \text{U}(d)$ . Let*

$$\Pi_{n,p}(\mu, \nu) = P_{n,p}(\mu) K_n(\nu \mid \mu), \quad Q_{m_n,p}(\nu) = \sum_{\mu} \Pi_{n,p}(\mu, \nu), \quad (20)$$

and, for  $\nu \in \mathbf{Y}_{m_n}$ , let

$$A_{n,p,\nu} = \sum_{\mu \in \mathbf{Y}_n} \Pi_{n,p}(\mu, \nu) \mathcal{E}_{\mu,\nu}(\rho_{p,\mu}) \quad (21)$$

be the unnormalized irreducible output block, so that  $\text{Tr } A_{n,p,\nu} = Q_{m_n,p}(\nu)$ . The fidelity of the lifted channel is given exactly by

$$F(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \sum_{\nu \in \mathbf{Y}_{m_n}} \sqrt{P_{m_n,p}(\nu)} F(A_{n,p,\nu}, \rho_{p,\nu}). \quad (22)$$

*Proof.* By covariance of  $\mathcal{Q}_n[K_n]$  and unitary invariance of fidelity, we may take  $g = I$ . Under the output Schur decomposition, the actual and target states are

$$\bigoplus_{\nu \in \mathbf{Y}_{m_n}} A_{n,p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad \bigoplus_{\nu \in \mathbf{Y}_{m_n}} P_{m_n,p}(\nu) \rho_{p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}.$$

Fidelity is additive over orthogonal direct sums, multiplicative over tensor products, and satisfies  $F(A, cB) = \sqrt{c} F(A, B)$  for  $c \geq 0$ . Since  $F(I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu, I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu) = 1$ , this proves (22).  $\square$

We now specialize the lifted channel by choosing its output-label kernel. The known- and unknown-spectrum protocols differ only in this choice: the former samples the exact target law, whereas the latter uses a randomized dilation of the measured input label.

## 2.2 Known spectrum: exact target-label sampling

Fix  $p \in \Delta_d^\circ$  and define

$$K_n^p(\nu \mid \mu) = P_{m_n, p}(\nu). \quad (23)$$

Thus  $\nu$  is independent of  $\mu$  and has the exact target law. Set

$$\mathcal{C}_n^p := \mathcal{Q}_n[K_n^p]. \quad (24)$$

The replacement branch in (19) leaves the sampled label unchanged, so the output Young law on  $\rho_{p, g}^{\otimes n}$  is  $P_{m_n, p}$  for every  $n$  and  $g$ .

## 2.3 Unknown spectrum: randomized rounding

For  $N \in \mathbb{N}$ , let

$$\mathbb{L}_N = \left\{ x \in \mathbb{Z}^d : \sum_i x_i = N \right\}.$$

Set  $\nu_n^{\text{fb}} = (m_n, 0, \dots, 0)$ . Given  $\mu \in \mathbb{Y}_n$ , draw independent variables  $U_1, \dots, U_{d-1}$ , each uniform on  $[-1/2, 1/2)$ , and set

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + U_i) + \frac{1}{2} \right\rfloor \quad (1 \leq i < d), \quad \tilde{\nu}_d = m_n - \sum_{i < d} \tilde{\nu}_i. \quad (25)$$

This is a spectrum-independent lattice version of the dilation  $\mu \mapsto \gamma_n \mu$ : it sends typical labels near  $np$  to labels near  $m_n p$ , with  $\tilde{\nu} - \mu$  typically a partition. The uniform shifts randomize the rounding error, avoiding lattice bias in the Gaussian limit. Then  $\tilde{\nu} \in \mathbb{L}_{m_n}$ . Let  $D_n(\tilde{\nu} \mid \mu)$  denote its distribution. Retain the rounded label when compatible, and otherwise use the fallback:

$$\nu = \begin{cases} \tilde{\nu}, & \tilde{\nu} - \mu \vdash m_n - n, \\ \nu_n^{\text{fb}}, & \text{otherwise.} \end{cases} \quad (26)$$

Compatibility implies  $\tilde{\nu} \in \mathbb{Y}_{m_n}$ , so the conditional law of  $\nu$  defines a Markov kernel  $K_n^{\text{rnd}} : \mathbb{Y}_n \rightsquigarrow \mathbb{Y}_{m_n}$ . Define

$$\mathcal{C}_n^{\text{univ}} := \mathcal{Q}_n[K_n^{\text{rnd}}]. \quad (27)$$

Since the kernel is independent of  $p$ , this is a spectrum-independent,  $U(d)$ -covariant cloner.

## 2.4 Sector and label asymptotics

The following proposition gives the common sector limit, including the uniformity under mixing needed when several input labels lead to one output label.

**Proposition 2.3** (Uniform Cartan-sector fidelity). *Let  $K \in \Delta_d^\circ$  and let  $\delta_n \geq 0$  satisfy  $\delta_n \rightarrow 0$ . For all sufficiently large  $n$ , every  $p \in K$ ,  $\mu \in \mathbb{Y}_n$ , and  $\nu \in \mathbb{Y}_{m_n}$  satisfying*

$$\left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \quad \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n$$

also satisfies  $\nu - \mu \vdash m_n - n$ . Moreover,

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) = F_{\text{orb}}(\gamma, p) + o(1) \quad (28)$$

uniformly over all such  $p, \mu, \nu$ . More generally, for fixed  $p, \nu$  as above and any probability distribution  $(a_\mu)_\mu$  supported on input labels satisfying the same closeness condition,

$$F\left(\sum_\mu a_\mu \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}\right) = F_{\text{orb}}(\gamma, p) + o(1), \quad (29)$$

uniformly over these distributions as well as  $p, \nu$ .

The mechanism is already visible on states obtained by a fixed number of lowering operations. Writing  $\omega = \nu - \mu$ , the Cartan intertwiner satisfies

$$V_{\mu,\omega} F_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)}) V_{\mu,\omega},$$

where  $F_\alpha^{(\lambda)}$  is the lowering operator for  $\alpha = e_i - e_j$  on  $\mathcal{H}_\lambda$ . After normalization, each excitation splits binomially between the two factors, with fraction

$$s_{\alpha,n} = \frac{\mu_i - \mu_j}{\nu_i - \nu_j} = \frac{1}{\gamma} + o(1)$$

entering the  $\mu$ -factor. Thus the conditional thermal parameter  $q_{ij} = p_j/p_i$  is transformed into

$$1 - s_{\alpha,n} + s_{\alpha,n} q_{ij} \longrightarrow 1 - \frac{1 - q_{ij}}{\gamma} =: q_{ij}^{(\gamma)}. \quad (30)$$

Write  $\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|$  for the thermal state in the number basis on the one-mode Fock space of [Section 3.2](#), and put

$$\tau_p = \bigotimes_{i < j} \tau_{q_{ij}}, \quad \tau_p^{(\gamma)} = \bigotimes_{i < j} \tau_{q_{ij}^{(\gamma)}}. \quad (31)$$

The low-weight matrices of the sector output and ideal sector therefore approach those of  $\tau_p^{(\gamma)}$  and  $\tau_p$ , whose fidelity is  $F_{\text{orb}}(\gamma, p)$ . The output coordinates depend on  $\nu$ , not on  $\mu$ , so the same state limit holds uniformly under mixing. This is a statement about conditional states for typical label pairs, not convergence of the varying channels  $\mathcal{C}_{\mu,\nu}$  to the Gaussian amplifier. The exact PBW calculation and uniform thermal-tail argument that complete the proof are given in [Appendix B](#).

The second estimate concerns only  $K_n^{\text{rnd}}$ .

**Proposition 2.4** (Randomized Young-label dilation). *There is a constant  $c_{\text{rnd}} < \infty$ , depending only on  $d$  and the fixed sequence  $(\gamma_n)$ , such that, for every compact  $K \Subset \Delta_d^\circ$  and all sufficiently large  $n$ , every  $p \in K$  and  $\mu \in \mathbb{T}_n(p)$  satisfy*

$$\text{supp } K_n^{\text{rnd}}(\cdot \mid \mu) \subseteq \{\nu \in \mathbb{Y}_{m_n} : \nu - \mu \vdash m_n - n, \quad \|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}\}. \quad (32)$$

Let

$$Q_{m_n,p}(\nu) := \sum_{\mu \in \mathbb{Y}_n} P_{n,p}(\mu) K_n^{\text{rnd}}(\nu \mid \mu), \quad \nu \in \mathbb{Y}_{m_n},$$

be the induced output-label law. Then, for every compact  $K \Subset \Delta_d^\circ$ ,

$$F(Q_{m_n,p}, P_{m_n,p}) = F_{\text{cl}}(\gamma, d) + o(1) \quad \text{uniformly for } p \in K. \quad (33)$$

The uniform dither in (25) makes the label kernel a discrete realization of continuous dilation. After centering at  $Np$  and scaling by  $\sqrt{N}$ , embed a label law as a piecewise-constant density on the fundamental cells of the label lattice. Before fallback replacement, the intersection-volume identity (147) identifies the embedded rounded law with the target-cell average of the  $\sqrt{\gamma_n}$ -dilation of the embedded input law. Write

$$\Sigma_p := (\text{diag}(p) - pp^\top) \Big|_{\{x \in \mathbb{R}^d: \sum_i x_i = 0\}}. \quad (34)$$

The Young local limit is  $\mathbf{N}_{0,\Sigma_p}$ , so randomized rounding changes the limit to  $\mathbf{N}_{0,\gamma\Sigma_p}$ , whereas the ideal  $m_n$ -label law still converges to  $\mathbf{N}_{0,\Sigma_p}$ . Their fidelity (Hellinger affinity) is

$$F(\mathbf{N}_{0,\gamma\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = \left( \frac{2\sqrt{\gamma}}{1+\gamma} \right)^{(d-1)/2} = F_{\text{cl}}(\gamma, d). \quad (35)$$

For typical input labels, every rounded output is compatible for all sufficiently large  $n$ ; fallback therefore changes only atypical inputs, whose total probability vanishes uniformly on compact spectral sets.

The proof of Proposition 2.4 is given in Appendix C.

## 2.5 Fidelity of the constructed cloners

We use the following continuity estimate for positive trace-class operators  $A, C, A', C'$  with trace at most one:

$$|F(A, C) - F(A', C')| \leq \|A - A'\|_1^{1/2} + \|C - C'\|_1^{1/2}. \quad (36)$$

Indeed, the Powers–Størmer inequality bounds  $\|\sqrt{A} - \sqrt{A'}\|_2$  by  $\|A - A'\|_1^{1/2}$ , and similarly for  $C, C'$ . Adding and subtracting  $\sqrt{A'}\sqrt{C}$ , the reverse triangle inequality and Schatten Hölder give (36), since  $\|\sqrt{C}\|_2, \|\sqrt{A'}\|_2 \leq 1$ .

The sector-mixture estimate gives a common factorization for both cloners.

**Lemma 2.5** (Factorization of the lifted fidelity). *Let  $K_n : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$  be label kernels. Fix  $K \in \Delta_d^\circ$ , and use the laws  $\Pi_{n,p}, Q_{m_n,p}$  from (20). Suppose there are sets  $G_{n,p}$  of label pairs and a sequence  $\delta_n \rightarrow 0$  such that*

$$\sup_{p \in K} \Pi_{n,p}(G_{n,p}^c) \rightarrow 0, \quad \max \left\{ \left\| \frac{\mu}{n} - p \right\|_\infty, \left\| \frac{\nu}{m_n} - p \right\|_\infty \right\} \leq \delta_n \quad ((\mu, \nu) \in G_{n,p}).$$

Then, uniformly for  $p \in K$  and  $g \in \mathbf{U}(d)$ ,

$$F\left(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{orb}}(\gamma, p)F(Q_{m_n,p}, P_{m_n,p}) + o(1). \quad (37)$$

*Proof.* By covariance, take  $g = I$ . Write  $P = P_{m_n,p}$ ,  $Q = Q_{m_n,p}$ , and  $G = G_{n,p}$ . Omitting the maximally mixed Specht factors, the output and target are  $O = \bigoplus_\nu A_{n,p,\nu}$  and  $T = \bigoplus_\nu P(\nu)\rho_{p,\nu}$ . For large  $n$ , all pairs in  $G$  are compatible by Proposition 2.3. Retain only these pairs:

$$O_G = \bigoplus_\nu A_\nu^G, \quad A_\nu^G = \sum_{\mu: (\mu, \nu) \in G} \Pi_{n,p}(\mu, \nu) \mathcal{C}_{\mu, \nu}(\rho_{p, \mu}), \quad Q_G(\nu) = \text{Tr } A_\nu^G.$$

Whenever  $Q_G(\nu) > 0$ , the mixture statement of Proposition 2.3 applies to  $A_\nu^G/Q_G(\nu)$ . The direct-sum fidelity identity (22) and  $F(Q_G, P) \leq 1$  therefore give

$$|F(O_G, T) - F_{\text{orb}}(\gamma, p)F(Q_G, P)| \leq \eta_n, \quad \eta_n \rightarrow 0,$$

uniformly on  $K$ . Positivity and trace preservation give  $\|O - O_G\|_1 = \|Q - Q_G\|_{\ell^1} = \Pi_{n,p}(G^c) =: \epsilon_{n,p}$ . Applying (36) once to the states and once to the label laws yields

$$|F(O, T) - F_{\text{orb}}(\gamma, p)F(Q, P)| \leq \eta_n + 2\sqrt{\epsilon_{n,p}} = o(1),$$

which proves (37).  $\square$

Now we can complete the achievability proofs of Theorems 1.1 and 1.2.

**Known spectrum.** For  $K_n^p$ , the joint law is  $P_{n,p} \otimes P_{m_n,p}$  and  $Q_{m_n,p} = P_{m_n,p}$ . Take  $G_{n,p} = \mathbb{T}_n(p) \times \mathbb{T}_{m_n}(p)$  and  $\delta_n = n^{-1/3} + m_n^{-1/3}$ . Young concentration makes the discarded mass vanish, while the label fidelity is exactly one. Thus (37), with  $K = \{p\}$ , gives

$$F(\mathcal{C}_n^p(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = F_{\text{orb}}(\gamma, p) + o(1),$$

proving achievability in Theorem 1.1.

**Unknown spectrum.** For  $K_n^{\text{rnd}}$ , take the pairs with  $\mu \in \mathbb{T}_n(p)$  and  $\nu \in \text{supp } K_n^{\text{rnd}}(\cdot | \mu)$ . The discarded mass is  $P_{n,p}(\mathbb{T}_n(p)^c) = o(1)$  uniformly on  $K$ , and (32) gives the required closeness with  $\delta_n = n^{-1/3} + c_{\text{rnd}}/m_n$ . Combining (37) and (33) yields

$$F(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) + o(1)$$

uniformly for  $p \in K$  and  $g \in \mathbb{U}(d)$ , proving the achievability assertion of Theorem 1.2.

### 3 Optimality from LAN and Gaussian converses

We prove the matching minimax upper bounds by restricting arbitrary cloners to local neighborhoods. Two-way LAN transports each cloner to an exact channel between Gaussian models, where dilation and amplification bounds apply. After defining the local coordinates, we establish these bounds, state LAN, and complete the minimax proofs. Figure 3 summarizes the transport; Lemma 3.5 proves the common fidelity estimate.

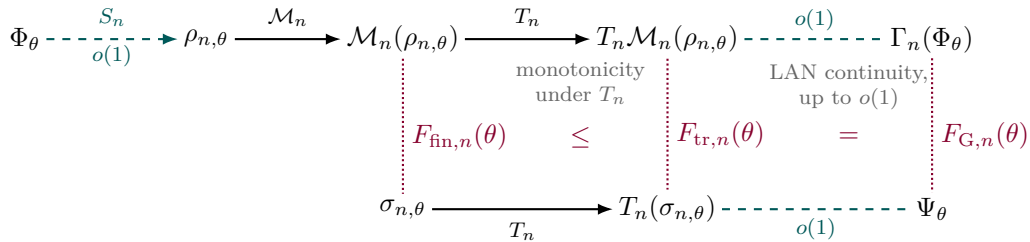


Figure 3: LAN transfer for  $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$  on a fixed compact local parameter set  $C$ . The top and bottom rows track output and target; dotted segments indicate fidelity pairs, solid arrows exact channels, and dashed links uniform  $o(1)$  trace-norm errors. Monotonicity and continuity give  $F_{\text{fin},n}(\theta) \leq F_{\text{tr},n}(\theta) = F_{\text{G},n}(\theta) + o(1)$  uniformly on  $C$ , as proved in Lemma 3.5. The LAN maps need not be inverses, and  $\Gamma_n$  need not be Gaussian or converge.



### 3.1 Local coordinates

Fix a base spectrum  $p \in \Delta_d^\circ$ . Let

$$\mathbf{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad Y_p(z) = \sum_{j < k} \frac{z_{jk} E_{kj} - \bar{z}_{jk} E_{jk}}{\sqrt{p_j - p_k}}, \quad z \in \mathbb{C}^{r_d},$$

where  $E_{jk}$  are the standard matrix units, and define

$$g_{n,z}^{(p)} = \exp\left(\frac{Y_p(z)}{\sqrt{n}}\right).$$

In  $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}$ , the coordinates  $h$  and  $z$  describe local changes of eigenvalues and eigenbasis, respectively. The denominators in  $Y_p$  use the fixed base spectrum  $p$ , including when the local spectrum is  $p + h/\sqrt{n}$ . Since  $Y_p(z)^* = -Y_p(z)$ , one has  $g_{n,z}^{(p)} \in \mathbf{U}(d)$ . For  $L > 0$ , set

$$\Omega_L = \{(h, z) \in \mathbf{H} \times \mathbb{C}^{r_d} : \|h\|_\infty \leq L, |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

For every fixed  $p \in \Delta_d^\circ$  and  $L > 0$ , the vector  $p + h/\sqrt{n}$  belongs to  $\Delta_d^\circ$  for every  $(h, z) \in \Omega_L$  once  $n$  is sufficiently large. Under LAN,  $h$  becomes a  $(d-1)$ -dimensional classical Gaussian shift, while  $z$  becomes a collection of displaced oscillator coordinates. The Gaussian theorem below optimizes the two corresponding fidelity losses jointly.

### 3.2 Gaussian amplification

Let  $\mathcal{F}$  be the one-mode Fock space with number basis  $\{|k\rangle : k \geq 0\}$ , creation and annihilation operators  $a^*, a$ , and number operator  $\hat{N} = a^*a$ . We use the Weyl convention

$$D(z) = \exp(za^* - \bar{z}a). \quad (38)$$

Recall the thermal state  $\tau_q = (1-q) \sum_{k \geq 0} q^k |k\rangle\langle k|$ ,  $0 < q < 1$ . Two thermal states commute, and hence

$$F(\tau_q, \tau_{q'}) = \frac{\sqrt{(1-q)(1-q')}}{1 - \sqrt{qq'}}. \quad (39)$$

For  $G \geq 1$ , put  $q^{(G)} = 1 - (1-q)/G$ . The quantum-limited phase-insensitive amplifier  $\mathcal{A}_G$  satisfies

$$\mathcal{A}_G(D(z)\tau_q D(z)^*) = D(\sqrt{G}z)\tau_{q^{(G)}} D(\sqrt{G}z)^*. \quad (40)$$

Equivalently,  $\mathcal{A}_G(X) = \text{Tr}_b[S_r(X \otimes |0\rangle\langle 0|_b)S_r^*]$ , where  $S_r$  is a two-mode squeezing unitary with  $\cosh^2 r = G$ . Squeezing supplies the gain  $\sqrt{G}$ , while tracing out the vacuum idler produces the minimum added noise encoded by  $q^{(G)}$ .

On  $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$ , use the thermal products  $\tau_p, \tau_p^{(\gamma)}$  from (31) and put  $D(z) = \bigotimes_{i < j} D(z_{ij})$ . The three relevant states are

$$\begin{aligned} \text{input:} \quad & \Phi_z^{(p)} = D(z)\tau_p D(z)^*, \\ \text{ideal gain-}\sqrt{\gamma}\text{ target:} \quad & \Psi_z^{(p,\gamma)} = D(\sqrt{\gamma}z)\tau_p D(\sqrt{\gamma}z)^*, \\ \text{amplifier output:} \quad & \mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}) = D(\sqrt{\gamma}z)\tau_p^{(\gamma)} D(\sqrt{\gamma}z)^*. \end{aligned} \quad (41)$$

Thus the ideal target retains the thermal state  $\tau_p$ ; only the actual amplifier output contains the additional noise  $\tau_p^{(\gamma)}$ . By (39),

$$F(\tau_{q^{(\gamma)}}, \tau_q) = f_\gamma(q). \quad (42)$$

Thus, for every  $z \in \mathbb{C}^{r_d}$ , the product-amplifier output and the corresponding ideal state satisfy

$$F\left(\mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}\right) = F_{\text{orb}}(\gamma, p). \quad (43)$$

Equip  $\mathbf{H}$  with its induced Euclidean measure and recall

$$\Sigma_p = (\text{diag}(p) - pp^\top) \Big|_{\mathbf{H}} > 0.$$

Write  $\mathbf{N}_{h, \Sigma_p}$  for the Gaussian law on  $\mathbf{H}$  with mean  $h$  and covariance  $\Sigma_p$ . A classical–quantum state has the form

$$R(x) = r(x)\rho_x, \quad r(x) \geq 0, \quad \int_{\mathbf{H}} r(x) \, dx = 1,$$

where  $x$  is a classical variable,  $r(x)$  is its density, and  $\rho_x$  is a quantum state on  $\mathcal{F}_{\text{orb}}$ . A channel  $\Lambda$  is completely positive (in the sense specified in Appendix D) and preserves  $\int \text{Tr } R(x) \, dx$ . If  $S(x) = s(x)\sigma_x$ , then

$$F(R, S) = \int_{\mathbf{H}} F(R(x), S(x)) \, dx = \int_{\mathbf{H}} \sqrt{r(x)s(x)} F(\rho_x, \sigma_x) \, dx.$$

With  $\|R\|_1 := \int_{\mathbf{H}} \|R(x)\|_1 \, dx$ , the usual monotonicity and joint-concavity properties remain valid, and the first equality above is the continuous direct-sum identity for fidelity.

Set

$$\Theta_{h,z}^{\text{in}} = \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)}, \quad \Theta_{h,z}^{\text{tar}} = \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p,\gamma)}. \quad (44)$$

Let

$$Z_L = \{z \in \mathbb{C}^{r_d} : |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

**Theorem 3.1** (Gaussian amplification optimum). *For every  $p \in \Delta_d^\circ$ ,*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F\left(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}\right) \, dz = F_{\text{orb}}(\gamma, p), \quad (45)$$

and

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F\left(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}\right) \, dh \, dz = F_{\text{univ}}(\gamma, p). \quad (46)$$

*The first supremum is over all quantum channels on  $\mathcal{F}_{\text{orb}}$ ; the second is over all classical–quantum channels. No Gaussianity, covariance, or product-structure assumption is imposed on the competing channels in either supremum.*

The achievability mechanism is explicit. The product amplifier  $\mathcal{A}_\gamma^{\otimes r_d}$  attains  $F_{\text{orb}}(\gamma, p)$  in the orbital model. In the full model, tensor it with the classical pushforward  $x \mapsto \sqrt{\gamma}x$ . The resulting Gaussian affinity equals the classical factor  $F_{\text{cl}}(\gamma, d)$ , so multiplicativity gives  $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) = F_{\text{univ}}(\gamma, p)$ .

For the converse, average a near-optimal channel over translates of the expanding flat-prior boxes. A weighted-fidelity witness reduces the problem to an oscillator moment. The corresponding weighted oscillator maps have a subsequential limit, tested against compact observables, that is completely positive and displacement covariant. Its output trace on normalized inputs is bounded by 1 in the orbital model and, by a direct Gaussian translation estimate, by  $F_{\text{cl}}(\gamma, d)$  in the full model. The least-noise inequality then gives the respective bounds  $F_{\text{orb}}(\gamma, p)$  and  $F_{\text{univ}}(\gamma, p)$ . The compactness and witness arguments are given in Proposition D.4 and Appendix D.

### 3.3 Local asymptotic normality

Here LAN means that, uniformly for  $(h, z)$  in compact sets, the state families  $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n}$  and  $\mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)}$  are asymptotically interconvertible by channels with vanishing trace-norm error. The Gaussian and oscillator factors represent spectral and eigenbasis fluctuations, respectively; cloning scales  $(h, z)$  by  $\sqrt{\gamma}$ .

We fix the oscillator normalization directly from the local tangent and the collective fluctuation operators. Put  $\Delta_{ij} = p_i - p_j$ . The chart in Section 3.1 satisfies

$$\frac{[Y_p(z), \rho_p]_{ji}}{\sqrt{\Delta_{ij}}} = z_{ij}, \quad i < j. \quad (47)$$

For  $a_{ij,n} = (n\Delta_{ij})^{-1/2} \sum_{s=1}^n E_{ij}^{(s)}$ , where  $E_{ij}^{(s)}$  acts on the  $s$ -th tensor factor, the expectations at  $\rho_p^{\otimes n}$  obey

$$\langle [a_{ij,n}, a_{ij,n}^*] \rangle = 1, \quad \langle a_{ij,n}^* a_{ij,n} \rangle = \frac{p_j}{\Delta_{ij}} = \frac{q_{ij}}{1 - q_{ij}}.$$

Under the local state, (47) gives  $\langle a_{ij,n} \rangle = z_{ij} + o(1)$ , uniformly on compact parameter sets. Thus the limiting mode has mean  $z_{ij}$  and thermal parameter  $q_{ij}$ , agreeing with  $D(z_{ij})\tau_{q_{ij}}D(z_{ij})^*$  in (38). This is also the rotation-to-displacement normalization in the constructive proof of [KG09, Lemma 6.4 and Section 7.3]. The following is Kahn–Guță’s LAN theorem [KG09, Theorem 4.3], expressed in our coordinates and supplemented by the target-sample rescaling below.

**Theorem 3.2** (Finite-dimensional LAN and target-sample scaling). *Fix  $p \in \Delta_d^\circ$ . There are channels*

$$T_N^{\text{full}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \longrightarrow L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})), \quad S_N^{\text{full}} : L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})) \longrightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, for every compact  $C \subset \mathbf{H} \times \mathbb{C}^{r_d}$ ,

$$\sup_{(h,z) \in C} \left\| T_n^{\text{full}} \left( \rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right) - \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right\|_1 \longrightarrow 0, \quad (48)$$

$$\sup_{(h,z) \in C} \left\| S_n^{\text{full}} \left( \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right) - \rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right\|_1 \longrightarrow 0, \quad (49)$$

and

$$\sup_{(h,z) \in C} \left\| T_{m_n}^{\text{full}} \left( \rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes m_n} \right) - \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p, \gamma)} \right\|_1 \longrightarrow 0. \quad (50)$$

**Coordinate identification and target scaling.** For (48)–(49), identify the spectral coordinate  $u \in \mathbb{R}^{d-1}$  of the cited theorem with  $h = Au = (u_1, \dots, u_{d-1}, -\sum_{i < d} u_i) \in \mathbf{H}$ . Its covariance  $V(p)_{ij} = \delta_{ij}p_i - p_i p_j$ ,  $i, j < d$  [KG09, Eq. (3.3)], satisfies  $AV(p)A^\top = \text{diag}(p) - pp^\top$ , giving  $\mathbf{N}_{h, \Sigma_p}$ ; the normalization (47) identifies the oscillator factor with  $\Phi_z^{(p)}$ .

For (50), write  $\gamma_n = m_n/n$  and use

$$\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}} = \rho_{p+\sqrt{\gamma_n}h/\sqrt{m_n}, g_{m_n, \sqrt{\gamma_n}z}^{(p)}}$$

with the same base  $p$  at both sample sizes. Apply (48) at sample size  $m_n$  on a common compact set containing  $(\sqrt{\gamma_n}h, \sqrt{\gamma_n}z)$  for all sufficiently large  $n$ . Since  $\gamma_n \rightarrow \gamma$ , uniform trace-norm continuity of Gaussian translations and Weyl conjugation on that set gives the claimed target limit.

*Remark 3.3* (Qubit normalization check). For  $d = 2$ , put  $\Delta = p_1 - p_2 > 0$ ,  $q = p_2/p_1$ , and take  $z = t \in \mathbb{R}$ . The exact qubit fidelity identity gives

$$F\left(\rho_p^{\otimes n}, \rho_{p, g_{n,t}}^{\otimes n}\right) = \left(1 - \Delta^2 \sin^2 \frac{t}{\sqrt{n\Delta}}\right)^{n/2} \longrightarrow e^{-\Delta t^2/2}.$$

This equals  $F(\tau_q, D(t)\tau_q D(t)^*) = \exp[-(1-q)t^2/(2(1+q))]$ , confirming the displacement scale in [Theorem 3.2](#).

**Corollary 3.4** (Fixed-spectrum LAN). *Fix  $p \in \Delta_d^\circ$ . There are channels*

$$T_N^{\text{orb}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \rightarrow \mathcal{T}_1(\mathcal{F}_{\text{orb}}), \quad S_N^{\text{orb}} : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \rightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

*such that, uniformly for  $z$  in compact sets,*

$$\|T_n^{\text{orb}}(\rho_{p, g_{n,z}}^{\otimes n}) - \Phi_z^{(p)}\|_1 \longrightarrow 0, \quad (51)$$

$$\|S_n^{\text{orb}}(\Phi_z^{(p)}) - \rho_{p, g_{n,z}}^{\otimes n}\|_1 \longrightarrow 0, \quad (52)$$

*and*

$$\|T_{m_n}^{\text{orb}}(\rho_{p, g_{n,z}}^{\otimes m_n}) - \Psi_z^{(p,\gamma)}\|_1 \longrightarrow 0. \quad (53)$$

*Proof.* Restrict [Theorem 3.2](#) to  $h = 0$ . The classical Gaussian component  $\mathbf{N}_{0, \Sigma_p}$  is independent of  $z$ . Set  $T_N^{\text{orb}}(X) := \int_{\mathbf{H}} T_N^{\text{full}}(X)(x) dx$  and  $S_N^{\text{orb}}(\sigma) := S_N^{\text{full}}(\mathbf{N}_{0, \Sigma_p} \otimes \sigma)$ . The bounds  $\|\int_{\mathbf{H}} R(x) dx\|_1 \leq \int_{\mathbf{H}} \|R(x)\|_1 dx$  and  $\|\mathbf{N}_{0, \Sigma_p} \otimes X\|_1 = \|X\|_1$  give the three assertions.  $\square$

The following lemma formalizes the transfer summarized in [Fig. 3](#).

**Lemma 3.5** (LAN fidelity transfer). *Let  $\{\rho_{n,\theta}\}_{\theta \in C}$  and  $\{\sigma_{n,\theta}\}_{\theta \in C}$  be input and target models on a compact parameter set  $C$ . Let  $\{\Phi_\theta\}$  and  $\{\Psi_\theta\}$  be limiting models, and suppose channels  $S_n, T_n$  satisfy*

$$\begin{aligned} \delta_n &:= \sup_{\theta \in C} \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \rightarrow 0, \\ \eta_n &:= \sup_{\theta \in C} \|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \rightarrow 0. \end{aligned}$$

*For every channel  $\mathcal{M}_n$ , put  $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$ . Then, for every  $\theta \in C$ ,*

$$F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) \leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}. \quad (54)$$

*The error is uniform over both  $\theta \in C$  and  $\mathcal{M}_n$ .*

*Proof.* Trace-norm contractivity of  $T_n \circ \mathcal{M}_n$  gives  $\|\Gamma_n(\Phi_\theta) - T_n \mathcal{M}_n(\rho_{n,\theta})\|_1 \leq \delta_n$ . Monotonicity under  $T_n$ , followed by [\(36\)](#) and the definition of  $\eta_n$ , therefore yields

$$F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) \leq F(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) \leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}.$$

$\square$

**Fixed-label asymptotics and their relation to LAN.** [Proposition 2.3](#) analyzes the quantum states for fixed input and output Young labels, before averaging over the label distributions. For label sequences with  $\mu_n/n \rightarrow p$  and  $\nu_n/m_n \rightarrow p$ , the target and cloner output, expressed in oscillator coordinates, approach  $\tau_p$  and  $\tau_p^{(\gamma)}$ , respectively, on each fixed excitation cutoff; see [\(144\)](#) and [\(143\)](#). The appearance of the same thermal state in LAN does not make these the same limit: LAN describes an entire local statistical model and provides channels connecting that model to its Gaussian approximation. Full LAN retains the Young-label fluctuation as  $N_{h,\Sigma_p}$ , whereas fixed-spectrum LAN sets  $h = 0$  and marginalizes this classical component. Thus the fixed-label calculation explains the achievable orbital factor  $F_{\text{orb}}(\gamma, p)$ , whereas LAN supplies the model for the converse.

### 3.4 Completion of the minimax proofs

We apply the transfer lemma to the orbital and full Gaussian models to complete [Theorems 1.1](#) and [1.2](#), respectively.

*Known-spectrum converse and completion of [Theorem 1.1](#).* For the known-spectrum problem, set  $u_n(p) := \sup_{\mathcal{M}_n} \inf_{g \in U(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$ . For every channel  $\mathcal{M}_n$  and fixed  $L$ , set

$$\Gamma_n = T_{m_n}^{\text{orb}} \circ \mathcal{M}_n \circ S_n^{\text{orb}}.$$

The local orbit  $\{g_{n,z}^{(p)} : z \in Z_L\}$  is contained in the full unitary orbit. Apply [Lemma 3.5](#), average over  $Z_L$ , and take the supremum over  $\mathcal{M}_n$ . Since the transfer error is independent of  $\mathcal{M}_n$ ,

$$u_n(p) \leq \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}) dz + o(1),$$

where the supremum is over all channels on the Gaussian model and the error tends to zero for fixed  $L$ . Taking  $\limsup_n$ , then  $L \rightarrow \infty$ , [\(45\)](#) gives

$$\limsup_{n \rightarrow \infty} u_n(p) \leq F_{\text{orb}}(\gamma, p).$$

Together with  $\liminf_n u_n(p) \geq F_{\text{orb}}(\gamma, p)$ , proved in [Section 2](#), this proves [Theorem 1.1](#).  $\square$

Fixing  $p$  in advance gives the known-spectrum problem and therefore cannot reveal the cost of an unknown spectrum. To capture that cost, we consider  $n^{-1/2}$ -neighborhoods of a fixed interior spectrum and let the local window grow after the large- $n$  limit. Each fixed local window eventually lies in the prescribed spectral set around any interior point, so the full Gaussian bound applies directly to the compact-set converse.

*Proof of part (b) of [Theorem 1.2](#).* For the compact-set problem, set  $w_n(K) := \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in U(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$ . Part (a) of [Theorem 1.2](#) gives

$$\liminf_{n \rightarrow \infty} w_n(K) \geq \inf_{p \in K} F_{\text{univ}}(\gamma, p).$$

For the reverse inequality, fix  $p_0 \in \text{int}_{A_d} K$ . For every fixed  $L$ , all local spectra  $p_0 + h/\sqrt{n}$ ,  $(h, z) \in \Omega_L$ , lie in  $K$  for sufficiently large  $n$ . For every channel  $\mathcal{M}_n$ , form

$$\Gamma_n = T_{m_n}^{\text{full}} \circ \mathcal{M}_n \circ S_n^{\text{full}},$$

using the LAN maps at  $p_0$ . The worst-case fidelity of  $\mathcal{M}_n$  on  $K$  is bounded by its average over this local window. The pointwise transfer bound, uniform over  $\mathcal{M}_n$ , therefore gives

$$w_n(K) \leq \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F\left(\Lambda(\Theta_{h,z}^{\text{in}}, \Theta_{h,z}^{\text{tar}})\right) dh dz + o(1),$$

where both Gaussian models are based at  $p_0$ , the supremum is over all classical–quantum channels, and  $L$  is fixed as  $n \rightarrow \infty$ . Taking  $\limsup_n$  and then  $L \rightarrow \infty$ , (46) yields

$$\limsup_{n \rightarrow \infty} w_n(K) \leq F_{\text{univ}}(\gamma, p_0).$$

Take the infimum over  $\text{int}_{\mathbf{A}_d} K$ . Its closure in  $\mathbf{A}_d$  is  $K$ , and  $F_{\text{univ}}(\gamma, \cdot)$  is continuous, so

$$\limsup_{n \rightarrow \infty} w_n(K) \leq \inf_{p \in K} F_{\text{univ}}(\gamma, p).$$

□

## 4 Cloning projector states

A flat rank- $r$  projector state has the form  $\rho_P = P/r$ , where  $P$  is a rank- $r$  orthogonal projector. These projectors form the complex Grassmannian  $\text{Gr}(r, d)$ . When  $r > 1$ , the state  $\rho_P$  has a nonzero eigenvalue of multiplicity  $r$ , so the simple-spectrum theorems do not apply. We construct  $\mathcal{C}_n^{\text{proj}}$  and prove Theorem 1.3 directly, without a general rank-deficient LAN theorem. The exact compression lemma below holds for any positive spectrum on its support, although we use it only for the flat spectrum. By itself, the lemma does not establish optimal cloning for other rank-deficient states.

### 4.1 Exact rank compression

In this section, a superscript  $(s)$  indicates the ambient dimension of a representation space or map. For a partition  $\lambda$  with at most  $r$  rows, the Weyl dimension formula gives

$$D_s^\lambda = \dim \mathcal{H}_\lambda^{(s)}, \quad R_\lambda = \frac{D_d^\lambda}{D_r^\lambda} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\lambda_i + j - i}{j - i}. \quad (55)$$

Let

$$J_\lambda : \mathcal{H}_\lambda^{(r)} \longrightarrow \mathcal{H}_\lambda^{(d)}$$

be the isometric  $U(r)$ -intertwiner that maps the unit highest-weight vector to the unit highest-weight vector. Its range is the  $U(r)$ -submodule generated by that vector, and we set  $P_\lambda = J_\lambda J_\lambda^*$ .

For a compatible pair of  $r$ -row partitions  $\mu \vdash n$  and  $\nu \vdash m$ , write  $\omega = \nu - \mu \vdash m - n$ . In ambient dimension  $s \in \{r, d\}$ , the Cartan-sector channel is

$$\mathcal{C}_{\mu, \nu}^{(s)}(X) = \frac{D_s^\mu}{D_s^\nu} (V_{\mu, \omega}^{(s)})^* (X \otimes I_{\mathcal{H}_\omega^{(s)}}) V_{\mu, \omega}^{(s)}.$$

For a probability vector  $x \in \mathbb{R}^s$ , including one with zero entries, write  $\pi_\lambda^{(s)}$  for the polynomial  $U(s)$ -representation,  $s_\lambda^{(s)}$  for its Schur polynomial, and

$$\rho_{x, \lambda}^{(s)} = \frac{\pi_\lambda^{(s)}(\text{diag } x)}{s_\lambda^{(s)}(x)}$$

whenever the denominator is nonzero.

**Lemma 4.1** (Exact rank-compression identity). *Let  $p = (p_1, \dots, p_r)$  satisfy  $p_i > 0$  and  $\sum_i p_i = 1$ , and put  $\bar{p} = (p_1, \dots, p_r, 0, \dots, 0) \in \mathbb{R}^d$ . For every  $r$ -row partition  $\lambda$ ,*

$$\rho_{\bar{p}, \lambda}^{(d)} = J_\lambda \rho_{p, \lambda}^{(r)} J_\lambda^*. \quad (56)$$

*For every compatible  $r$ -row pair  $\mu, \nu$  and every trace-class operator  $X$  on  $\mathcal{H}_\mu^{(r)}$ ,*

$$P_\nu \mathcal{C}_{\mu, \nu}^{(d)}(J_\mu X J_\mu^*) P_\nu = a_{\mu, \nu}^{(r, d)} J_\nu \mathcal{C}_{\mu, \nu}^{(r)}(X) J_\nu^*, \quad (57)$$

*where*

$$a_{\mu, \nu}^{(r, d)} = \frac{R_\mu}{R_\nu} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\mu_i + j - i}{\nu_i + j - i}. \quad (58)$$

*Consequently, the fidelity for each pair of labels factorizes exactly:*

$$F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{\bar{p}, \mu}^{(d)}), \rho_{\bar{p}, \nu}^{(d)}\right) = \sqrt{a_{\mu, \nu}^{(r, d)}} F\left(\mathcal{C}_{\mu, \nu}^{(r)}(\rho_{p, \mu}^{(r)}), \rho_{p, \nu}^{(r)}\right). \quad (59)$$

*Proof.* The singular specialization identity for Schur polynomials is

$$s_\lambda^{(d)}(p_1, \dots, p_r, 0, \dots, 0) = s_\lambda^{(r)}(p_1, \dots, p_r),$$

and the left side vanishes when  $\lambda$  has more than  $r$  rows. We claim that

$$\pi_\lambda^{(d)}(\text{diag } \bar{p}) = J_\lambda \pi_\lambda^{(r)}(\text{diag } p) J_\lambda^*.$$

Indeed,  $\pi_\lambda^{(d)}(\text{diag } \bar{p})$  acts on each weight space of  $\mathcal{H}_\lambda^{(d)}$  by the monomial  $\bar{p}^\xi$ , which vanishes unless the weight  $\xi$  is supported in the first  $r$  coordinates. By the Gelfand–Tsetlin branching rule, the weight spaces so supported span exactly the  $U(r)$ -submodule generated by the highest-weight vector, that is, the range of  $J_\lambda$ ; on that submodule the block-diagonal action reduces to the  $r$ -dimensional polynomial representation, proving the claim. Normalizing by the Schur-polynomial identity proves (56).

The two maps

$$V_{\mu, \omega}^{(d)} J_\nu, \quad (J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}$$

are  $U(r)$ -intertwiners from  $\mathcal{H}_\nu^{(r)}$  to  $\mathcal{H}_\mu^{(d)} \otimes \mathcal{H}_\omega^{(d)}$ . Both send the unit highest-weight vector to  $v_\mu \otimes v_\omega$ , so they agree on the submodule generated by that vector:

$$V_{\mu, \omega}^{(d)} J_\nu = (J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}. \quad (60)$$

Inserting (60) into the two sector-channel formulas proves (57), with the dimension ratio in (58); its product form follows from (55).

If  $A, B \geq 0$  and  $PBP = B$ , then

$$F(A, B) = \text{Tr} \sqrt{\sqrt{B} A \sqrt{B}} = F(PAP, B).$$

Apply this with  $P = P_\nu$ , then use (56), (57), invariance of fidelity under the isometry  $J_\nu$ , and  $F(aA, B) = \sqrt{a} F(A, B)$ . This proves (59). Compatibility gives  $\nu_i \geq \mu_i$ , so every factor in (58) is at most one.  $\square$

For the flat spectrum and the standard projector  $P_0$  onto the first  $r$  coordinates, the conditional state in the  $\lambda$ -sector is

$$\rho_{P_0, \lambda} = J_\lambda \frac{I_{\mathcal{H}_\lambda^{(r)}}}{D_r^\lambda} J_\lambda^*. \quad (61)$$

The  $r$ -dimensional Cartan channel is  $U(r)$ -covariant and therefore maps the maximally mixed state on the irreducible input module exactly to the maximally mixed state on the irreducible output module. The compression identity (59) consequently becomes, for every compatible pair,

$$F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{P_0, \mu}), \rho_{P_0, \nu}\right) = \sqrt{\frac{R_\mu}{R_\nu}}. \quad (62)$$

For  $\mu_i/n \rightarrow 1/r$  and  $\nu_i/m_n \rightarrow 1/r$ , this tends to the right-hand side of (9).

## 4.2 An asymptotically compatible coupling with exact marginals

Let

$$\mathbf{P}_N^{(r)}(\lambda) = \frac{\dim \mathcal{S}_\lambda D_r^\lambda}{r^N}, \quad \lambda \vdash N, \quad \ell(\lambda) \leq r, \quad (63)$$

be the Schur–Weyl law of the maximally mixed state on  $\mathbb{C}^r$ . The complete Schur decomposition is

$$(P/r)^{\otimes N} = \bigoplus_{\ell(\lambda) \leq r} \mathbf{P}_N^{(r)}(\lambda) \frac{\Pi_{\lambda, P}}{D_r^\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}. \quad (64)$$

Here  $\Pi_{\lambda, P}$  is the rank- $D_r^\lambda$  support projection defined intrinsically by the range of  $P$ . Padding partitions with zero rows to length  $r$ , the Schur–Weyl/random-word central limit theorem [Kup02, Joh01] states that

$$X_N := \frac{\lambda - (N/r)\mathbf{1}_r}{\sqrt{N}} \Rightarrow X, \quad (65)$$

where  $X$  is a fixed scaling of the ordered eigenvalue vector of a traceless  $r \times r$  GUE matrix. In particular,  $\Pr(X_1 > \dots > X_r) = 1$ .

Both  $X_n$  and  $X_{m_n}$  converge to the same law. Their Prokhorov distance therefore tends to zero, so Strassen’s coupling theorem [Str65] supplies a coupling  $\Pi_n(\mu, \nu)$  of the exact marginals  $\mathbf{P}_n^{(r)}$  and  $\mathbf{P}_{m_n}^{(r)}$  such that

$$\left\| \frac{\mu - (n/r)\mathbf{1}_r}{\sqrt{n}} - \frac{\nu - (m_n/r)\mathbf{1}_r}{\sqrt{m_n}} \right\|_\infty \rightarrow 0 \quad \text{in probability.} \quad (66)$$

Disintegrate this finite coupling to obtain a label kernel  $K_n(\nu \mid \mu)$ , extending it arbitrarily to input diagrams with more than  $r$  rows. Those diagrams have zero probability on projector inputs, but the extension defines the lifted channel on the entire input space. Define the projector cloner by

$$\mathcal{C}_n^{\text{proj}} := \mathcal{Q}_n[K_n]. \quad (67)$$

Its dependence on the fixed rank  $r$  is suppressed in the notation; the channel is independent of the unknown projector  $P$ .

Put  $\omega = \nu - \mu$ . Its leading term is  $(m_n - n)/r$  in every coordinate, while its fluctuations are of order  $\sqrt{n}$ . Hence  $\omega_i > 0$  for every  $i$  with probability  $1 - o(1)$ . For adjacent rows,

$$\frac{\omega_i - \omega_{i+1}}{\sqrt{n}} = \sqrt{\gamma_n} \left[ \frac{\nu_i - \nu_{i+1}}{\sqrt{m_n}} \right] - \left[ \frac{\mu_i - \mu_{i+1}}{\sqrt{n}} \right] \Rightarrow (\sqrt{\gamma} - 1)(X_i - X_{i+1}) > 0.$$



Thus the compatibility event

$$C_n = \{(\mu, \nu) : \nu - \mu \vdash m_n - n\}$$

has probability  $1 - o(1)$ . The cloner  $\mathcal{C}_n^{\text{proj}}$  uses the ambient Cartan-sector channel on  $C_n$  and, otherwise, the covariant replacement channel into the same sampled  $\nu$ -block. The output-label marginal is exactly the target law. The central limit theorem also gives  $\mu_i/n, \nu_i/m_n \rightarrow 1/r$  in probability, so (55) implies

$$\mathbf{1}_{C_n} \sqrt{\frac{R_\mu}{R_\nu}} \xrightarrow{\mathbb{P}} \gamma^{-r(d-r)/2}. \quad (68)$$

On  $C_n$ , compatibility gives  $\nu_i \geq \mu_i$ , hence  $0 \leq \sqrt{R_\mu/R_\nu} \leq 1$ . The displayed random variables are therefore bounded by one, and their expectations converge to the same limit. As in the proof of Lemma 2.5, use the block decomposition and discard incompatible pairs. The block identity (22), valid verbatim for the  $r$ -row label laws, together with (62) and concavity of fidelity gives

$$F(\mathcal{C}_n^{\text{proj}}((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \geq \mathbb{E}_{\Pi_n} \left[ \mathbf{1}_{C_n} \sqrt{\frac{R_\mu}{R_\nu}} \right] \rightarrow \gamma^{-r(d-r)/2}. \quad (69)$$

This proves achievability in Theorem 1.3.

### 4.3 Direct finite-dimensional converse

Covariantize an arbitrary cloner over  $U(d)$  and twirl its output over permutations. Joint concavity and unitary invariance of fidelity justify the first step; monotonicity under the output twirl, which fixes every target, justifies the second. Neither decreases the worst-case fidelity over projector states. Write  $\mathcal{M}_n$  for the resulting channel.

Replace each support projection in (64) by the identity to obtain the invariant positive operator

$$B_n = \bigoplus_{\substack{\mu \vdash n \\ \ell(\mu) \leq r}} \mathbf{P}_n^{(r)}(\mu) \frac{I_{\mathcal{H}_\mu^{(d)}}}{D_r^\mu} \otimes \frac{I_{\mathcal{S}_\mu}}{\dim \mathcal{S}_\mu}.$$

For every  $P$ ,

$$(P/r)^{\otimes n} \leq B_n, \quad \text{Tr } B_n = \sum_{\mu} \mathbf{P}_n^{(r)}(\mu) R_\mu.$$

Covariance and the output twirl imply that

$$\mathcal{M}_n(B_n) = \bigoplus_{\substack{\nu \vdash m_n \\ \ell(\nu) \leq d}} z_\nu \frac{I_{\mathcal{H}_\nu^{(d)}}}{D_d^\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad z_\nu \geq 0, \quad \sum_{\nu} z_\nu = \text{Tr } B_n.$$

Fidelity is increasing in either positive argument, since  $A \leq B$  implies  $\text{Tr} \sqrt{\sqrt{T} A \sqrt{T}} \leq \text{Tr} \sqrt{\sqrt{T} B \sqrt{T}}$ . Positivity of  $\mathcal{M}_n$  and the direct-sum formula therefore give

$$\begin{aligned} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) &\leq F(\mathcal{M}_n(B_n), (P/r)^{\otimes m_n}) \\ &= \sum_{\ell(\nu) \leq r} \sqrt{\frac{z_\nu \mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu}}. \end{aligned}$$

By Cauchy–Schwarz and the trace identity for  $B_n$ ,

$$F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \leq \left[ \left( \sum_{\mu} \mathbf{P}_n^{(r)}(\mu) R_{\mu} \right) \left( \sum_{\nu} \frac{\mathbf{P}_{m_n}^{(r)}(\nu)}{R_{\nu}} \right) \right]^{1/2}. \quad (70)$$

Both sums run over partitions with at most  $r$  rows; output mass in longer diagrams can only strengthen the inequality.

For the asymptotics, put  $q = r(d - r)$ . Since  $\lambda_i/N \rightarrow 1/r$  in probability under  $\mathbf{P}_N^{(r)}$ , (55) gives

$$\frac{R_{\lambda}}{N^q} \xrightarrow{\mathbb{P}} c_{r,d} := \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{1}{r(j-i)} > 0.$$

The ratio  $R_{\lambda}/N^q$  is globally bounded, since  $\lambda_i \leq N$ . Its reciprocal is bounded on the fixed-cutoff event  $E_N = \{\lambda_r \geq N/(2r)\}$ , whereas globally  $N^q/R_{\lambda} \leq N^q$ , because  $R_{\lambda} \geq 1$ . The finite Young large-deviation estimate (16), applied in dimension  $r$ , and Pinsker’s inequality give  $\mathbf{P}_N^{(r)}(E_N^c) \leq (N+1)^{C_r} e^{-c_r N}$  for constants  $C_r, c_r > 0$ . Thus the reciprocal’s expectation on  $E_N^c$  satisfies  $\mathbb{E}[(N^q/R_{\lambda})\mathbf{1}_{E_N^c}] \rightarrow 0$ . Bounded convergence in probability on  $E_N$  therefore yields

$$\mathbb{E}_{\mathbf{P}_N^{(r)}} R_{\lambda} \sim c_{r,d} N^q, \quad \mathbb{E}_{\mathbf{P}_N^{(r)}} \frac{1}{R_{\lambda}} \sim c_{r,d}^{-1} N^{-q}. \quad (71)$$

Substitution into (70) yields

$$\limsup_{n \rightarrow \infty} F \leq \gamma^{-r(d-r)/2}, \quad (72)$$

which completes the converse and the proof of [Theorem 1.3](#).

## 5 Discussions

We first compare our cloners with PCT at fixed gain and in the small-error regime, then discuss conjectured extensions to degenerate and rank-deficient spectra and the remaining open problems.

### 5.1 Fixed-gain comparison with PCT

Let  $N$  be the number of inputs and  $M$  the number of *additional* outputs, so that

$$m = N + M, \quad \delta = \frac{M}{N}, \quad \gamma = 1 + \delta. \quad (73)$$

For a rank bound  $r$ , purify–clone–trace (PCT) applies random purification [TWZ25], Werner’s pure-state cloner in dimension  $dr$  [Wer98], and partial trace over the purifying registers [LTHC26, FGSS26, JSO26]. Denote its output on  $\rho^{\otimes N}$  by  $\sigma_{N,N+M}^{\text{PCT}}(\rho)$ . We report protocol comparisons in squared Uhlmann fidelity  $F^2$ , separating fixed-gain fidelity from small-error sample complexity.

Take  $N = n$ ,  $N + M = m_n$ , and  $m_n/n \rightarrow \gamma > 1$ , as illustrated in [Fig. 2](#). For every  $\rho$  of rank at most  $r$ , the published data-processing guarantee is

$$F\left(\sigma_{N,N+M}^{\text{PCT}}(\rho), \rho^{\otimes(N+M)}\right)^2 \geq \frac{\binom{N+dr-1}{dr-1}}{\binom{N+M+dr-1}{dr-1}} \longrightarrow \gamma^{-(dr-1)}. \quad (74)$$

The ratio is exact for each purified component before averaging and discarding the purifying registers; it is only a lower bound for the final output. [Appendix A](#) analyzes that output to establish the strict protocol comparisons below.

**Simple full-rank spectra.** Fix  $p_1 > \dots > p_d > 0$ . PCT uses rank bound  $r = d$  and purification dimension  $d^2$ ; neither it nor  $\mathcal{C}_n^{\text{univ}}$  is supplied  $p$ . Write  $\sigma_{n,m}^{\text{PCT}}(p)$  for its output on  $\rho_p^{\otimes n}$ . With  $q_{ij} = p_j/p_i$ , its exact asymptotic fidelity is

$$F_{\text{PCT}}(\gamma, p) = \left( \frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(d-1)/2} \prod_{i < j} \frac{\sqrt{\gamma + (\gamma - 1)q_{ij}} + \sqrt{q_{ij}(\gamma - 1 + \gamma q_{ij})}}{\gamma(1 + q_{ij})}. \quad (75)$$

By [Theorems A.2](#) and [1.2](#),

$$\lim_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}\right)^2 = F_{\text{PCT}}(\gamma, p)^2 < F_{\text{univ}}(\gamma, p)^2, \quad (76)$$

where the right-hand side is the exact limiting squared fidelity of our universal cloner. The derivation and strict comparison are proved in [Appendix A](#).

**Projector states.** For  $\rho = P/r$ ,  $P \in \text{Gr}(r, d)$ , and  $1 \leq r < d$ , the projector family has complex dimension  $r(d - r)$ . [Theorem 1.3](#) gives its exact limiting squared fidelity  $\gamma^{-r(d-r)}$ . For PCT with rank bound  $r$ , write  $\sigma_{n,m}^{\text{PCT}}(P)$  for the output on  $(P/r)^{\otimes n}$ . The appendix gives

$$\limsup_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n}\right)^2 \leq \gamma^{-r(d-r)} \left( \frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{r-1} < \gamma^{-r(d-r)} \quad (1 < r < d). \quad (77)$$

Thus our projector cloner  $\mathcal{C}_n^{\text{proj}}$  strictly improves on PCT for every fixed  $\gamma > 1$  and  $1 < r < d$ . For  $r = 1$ , both reduce to Werner's optimal pure-state cloner; for  $r = d$ , the projector family is the known singleton  $I_d/d$ .

**Worst-case cloning over all states.** At fixed gain  $\gamma > 1$ , we now allow the input to be any  $d$ -dimensional state, with no spectral or rank information supplied to the cloner. The best fidelity that a single channel can guarantee over this entire family is

$$w_n = \sup_{\mathcal{M}_n} \inf_{\substack{\rho \geq 0 \\ \text{Tr } \rho = 1}} F(\mathcal{M}_n(\rho^{\otimes n}), \rho^{\otimes m_n}). \quad (78)$$

The PCT guarantee gives  $\liminf_n w_n \geq \gamma^{-(d^2-1)/2}$ . Restricting to the compact simple-spectrum chambers  $\Delta_{d,\kappa}$ , applying [\(8\)](#), and then letting  $\kappa \downarrow 0$  gives

$$\limsup_{n \rightarrow \infty} w_n \leq \inf_{p \in \Delta_d^\circ} F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) \gamma^{-d(d-1)/4}. \quad (79)$$

For the equality, each orbital factor is at least  $f_\gamma(0) = \gamma^{-1/2}$ , and the geometric spectra  $p_i(t) = t^{i-1} / \sum_{j=1}^d t^{j-1}$ ,  $0 < t < 1$ , approach this bound as  $t \downarrow 0$ , since every ratio  $p_j(t)/p_i(t)$ ,  $i < j$ , tends to zero. At large gain, these lower and upper bounds scale respectively as  $\gamma^{-(d^2-1)/2}$  and  $\gamma^{-(d^2-1)/4}$ , up to constants, leaving a factor of two between the polynomial exponents. The exact all-state minimax value and whether PCT attains it remain open.

## 5.2 Small-error sample complexity

For  $1 \leq r \leq d$  and sufficiently small  $\epsilon > 0$ , the concurrent papers [\[FGSS26, JSO26\]](#) prove that cloning with squared fidelity at least  $1 - \epsilon$  for every state of rank at most  $r$  has optimal worst-case sample complexity

$$N = \Theta\left(\frac{Mrd}{\epsilon}\right). \quad (80)$$

Their upper bounds use PCT, while their lower bounds use flat projectors of rank  $\ell = \Theta(r)$ , possibly  $\ell < r$ . In particular, the rank bound  $r = d$  does not restrict the hard family to  $I_d/d$ . Our limiting fidelities give more precise comparisons on the specified spectral families, as follows.

**Simple full-rank spectra.** For fixed  $p$  and  $\bullet \in \{\text{univ}, \text{PCT}\}$ , expansion at  $\gamma = 1 + \delta$  and inversion at squared fidelity  $1 - \epsilon$  give

$$F_{\bullet}(1 + \delta, p)^2 = 1 - A_{\bullet}(p)\delta^2 + O_p(\delta^3), \quad \frac{N}{M} \sim \sqrt{\frac{A_{\bullet}(p)}{\epsilon}}, \quad (81)$$

where

$$A_{\text{univ}}(p) = \frac{d-1}{8} + \frac{1}{4} \sum_{i < j} \frac{p_i}{p_j}, \quad (82)$$

$$A_{\text{PCT}}(p) = \frac{d-1}{2} + \frac{1}{4} \sum_{i < j} \frac{(p_i + p_j)^2}{p_i p_j}. \quad (83)$$

Thus  $A_{\text{PCT}}(p) > A_{\text{univ}}(p)$ , giving our cloner the strictly smaller leading constant. Both coefficients diverge as an eigenvalue approaches zero, so this fixed-interior-spectrum  $1/\sqrt{\epsilon}$  law does not contradict the worst-case  $1/\epsilon$  scaling over the rank-bounded family.

**Projector states.** For  $1 < r < d$ , inverting our limiting squared fidelity gives

$$\frac{N}{M} = \frac{1}{(1 - \epsilon)^{-1/[r(d-r)]} - 1} \sim \frac{r(d-r)}{\epsilon}. \quad (84)$$

The published guarantee (74) certifies PCT squared fidelity at least  $1 - \epsilon$  whenever

$$\frac{N}{M} \geq \frac{1}{(1 - \epsilon)^{-1/(rd-1)} - 1} \sim \frac{rd-1}{\epsilon}, \quad (85)$$

since its binomial ratio is  $\prod_{j=1}^{rd-1} (N+j)/(N+M+j) \geq [N/(N+M)]^{rd-1}$ . The coefficient  $rd-1$  belongs only to this sufficient condition. In fact, Proposition A.4 in Appendix A shows that PCT's actual squared infidelity is  $r(d-r)\delta + O_{r,d}(\delta^2)$  in the iterated limit specified below, for both the asymptotic  $\liminf$  and  $\limsup$ . Thus PCT also has leading sample ratio  $N/M \sim r(d-r)/\epsilon$ : our strict fixed-gain advantage does not improve this leading coefficient. Together with (77), these bounds show that our advantage first appears at second order: the asymptotic squared-fidelity gap is of order  $\delta^2$  in this iterated sense, although its exact coefficient is not determined here. Subleading expansions of the optimal and certified thresholds are recorded in (126).

**Order of limits.** The strict protocol comparisons hold at every fixed  $\gamma > 1$ . The small-error thresholds derived from our limiting fidelities use the iterated limit  $N, M \rightarrow \infty$  with  $M/N \rightarrow \delta > 0$ , followed by  $\delta \downarrow 0$ . We have not established approximation-error bounds uniform as  $\gamma_n \downarrow 1$ , so these thresholds do not provide finite-sample bounds in a joint high-fidelity regime or improve the worst-case  $1/\epsilon$  scaling over the rank-bounded family up to constants. The sufficient condition (85) is a finite-sample guarantee and does not require this iterated limit.

### 5.3 Conjectures for degenerate and rank-deficient spectra

We record two expected extensions of the proved formulas, distinguishing known-spectrum orbits from local models with unknown spectrum and fixed rank. Suppose a rank- $r$  state  $\rho$  has distinct eigenvalues  $u_1 > \dots > u_s > 0$ , with multiplicities  $r_1, \dots, r_s$ , where  $\sum_a r_a = r$  and  $\sum_a r_a u_a = 1$ . We conjecture that the known-spectrum optimum on its unitary orbit, defined as in (6), is

$$F_{\text{orb}}(\gamma, \rho) = \gamma^{-r(d-r)/2} \prod_{a < b} f_\gamma\left(\frac{u_b}{u_a}\right)^{r_a r_b}. \quad (86)$$

Only pairs of distinct positive eigenspaces contribute to the product: rotations within a degenerate eigenspace are stabilizer directions, while the  $r(d-r)$  support–kernel directions give the first factor. The simple full-rank case  $r = d$ ,  $s = d$ , and  $r_a = 1$  reduces to the orbital formula in (5); the flat projector case  $s = 1$  reduces to  $\gamma^{-r(d-r)/2}$ , proved in Theorem 1.3. In particular,  $r = 1$  recovers Werner’s pure-state value  $\gamma^{-(d-1)/2}$ . The exact compression identity in Lemma 4.1 establishes this support–kernel factorization for our sector channel, but does not by itself prove the conjecture for the remaining spectra.

The known-spectrum optimum need not be continuous at a degeneracy, because the orbit loses dimensions. For example, for qubits,

$$\lim_{\substack{p \rightarrow (1/2, 1/2) \\ p_1 > p_2}} F_{\text{orb}}(\gamma, p) = \frac{2\sqrt{\gamma}}{1 + \gamma} < 1,$$

whereas the known-spectrum family at  $p = (1/2, 1/2)$  contains only  $I_2/2$  and can be cloned perfectly. At any fixed sample size, preparing  $(I_2/2)^{\otimes m_n}$  gives fidelity tending uniformly to one as  $p \rightarrow (1/2, 1/2)$ . Thus the degeneracy and large-sample limits do not commute.

For the local unknown-state problem, consider the smooth manifold of rank- $r$  states

$$\mathcal{D}_r^\circ := \{\sigma \succeq 0 : \text{Tr } \sigma = 1, \text{ rank } \sigma = r\}. \quad (87)$$

If  $P$  is the support projection of  $\rho$ , its tangent space is

$$T_\rho \mathcal{D}_r^\circ = \{X = X^* : \text{Tr } X = 0, (I - P)X(I - P) = 0\}. \quad (88)$$

Let  $\text{Exp}_\rho$  be the exponential map for the induced Hilbert–Schmidt metric. For fixed  $L$  and sufficiently large  $n$ , set  $\rho_{n,X} = \text{Exp}_\rho(X/\sqrt{n})$  and define

$$v_{n,L}^{(r)}(\rho) := \sup_{\mathcal{M}_n} \inf_{\substack{X \in T_\rho \mathcal{D}_r^\circ \\ \|X\|_2 \leq L}} F\left(\mathcal{M}_n(\rho_{n,X}^{\otimes n}), \rho_{n,X}^{\otimes m_n}\right). \quad (89)$$

Here  $\rho$  is fixed before the supremum, so the channel sequence may depend on the center  $\rho$ , but not on the unknown local displacement  $X$ . We conjecture that the  $\limsup$  and  $\liminf$  of  $v_{n,L}^{(r)}(\rho)$ , first as  $n \rightarrow \infty$  and then as  $L \rightarrow \infty$ , agree. Writing their common value as  $F_{\text{univ}}^{(r)}(\gamma, \rho)$ , the conjectured formula is

$$F_{\text{univ}}^{(r)}(\gamma, \rho) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma}\right)^{(\sum_{a=1}^s r_a^2 - 1)/2} F_{\text{orb}}(\gamma, \rho). \quad (90)$$

The classical exponent is half the real dimension  $\sum_a r_a^2 - 1$  of Hermitian matrices block diagonal in the positive eigenspaces and with zero total trace. The support–kernel directions are already included in  $F_{\text{orb}}(\gamma, \rho)$ , and the degenerate zero block contributes no additional parameters when the

rank is fixed. Gaussian matrix fluctuations suggest a classical limit for the within-block directions [Kup02, Joh01]. For simple positive spectra (every  $r_a = 1$ ), the fixed-rank LAN theorem of [LN24, Theorem 3.3] has exactly  $r - 1$  classical Gaussian parameters,  $r(r - 1)/2$  shifted thermal modes, and  $r(d - r)$  shifted pure modes, matching the exponent  $(r - 1)/2$  here and the factors in (86).

The proposed formula is also consistent under positive-eigenvalue collisions at fixed rank: each colliding pair contributes  $f_\gamma(1) = 2\sqrt{\gamma}/(1 + \gamma)$ , and  $(r - 1)/2 + \sum_a \binom{r_a}{2} = (\sum_a r_a^2 - 1)/2$ . When also  $r = d$  and the spectrum is simple, the formula reduces to the full-rank value  $F_{\text{univ}}(\gamma, p)$  in (5).

At a flat projector state,  $s = 1$ , the conjectured universal fidelity is different from Theorem 1.3 which is rank-aware and does not address neighborhoods in which the rank can change.

**Conjectured comparison with PCT.** At every fixed gain  $\gamma > 1$ , we further conjecture that our Schur–Cartan construction, with label kernels adapted to degenerate or rank-deficient spectra, generally outperforms PCT on the corresponding mixed-state families, with the same prior spectral or rank information available to both protocols. We also conjecture that one spectrum-independent extension of our construction achieves strictly larger asymptotic worst-case fidelity over all  $d$ -dimensional states than PCT with rank bound  $d$ , in the all-state minimax problem (78). We do not pursue these comparisons here; such a worst-case separation would require boundary control beyond the pointwise comparisons proved above.

## 5.4 Further open problems

Beyond these fixed-rank conjectures, uniform control near rank changes and the exact minimax fidelity over all states at fixed gain remain open. The upper bound (79) and the PCT lower guarantee leave a gap even between the large-gain exponents. The finite-copy optimum is also unknown: asymptotic optimality does not establish optimality of our channels at each finite  $n$ . Finally, approximation-error bounds controlled as  $\gamma_n \downarrow 1$ , on the scale of the vanishing infidelity, would be needed to turn the iterated small-error comparisons into joint high-fidelity bounds.

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## Generative AI usage

The original idea for the cloner construction is due to the authors. We expect its optimality in the asymptotic regime, but lack a complete proof. We used OpenAI’s ChatGPT (Sol 5.6 and 6) and Anthropic’s Claude (Fable 5 and 5.1) to assist with the analysis: the evaluation of the cloning fidelity and the converse argument benefited from their suggestions. The same tools assisted with language editing, manuscript organization, and figure preparation. All AI-assisted material was reviewed and revised by the authors, who take full responsibility for the accuracy and content of the manuscript.

## A Strict comparison with purify–clone–trace

We compute the exact asymptotic PCT fidelity at simple full-rank spectra and derive a strict upper bound for flat projector states with  $1 < r < d$ , proving the comparisons stated in [Section 5](#).

### A.1 Common preliminaries

For  $D \geq 1$ , write  $d_D(N) = \dim \text{Sym}^N(\mathbb{C}^D) = \binom{N+D-1}{D-1}$ , and let  $\Pi_{\text{sym},N}^{(D)}$  project onto this symmetric subspace. On inputs supported on  $\text{Sym}^n(\mathbb{C}^D)$ , Werner’s pure-state cloner [\[Wer98\]](#) is

$$\mathcal{W}_{n,m}^{(D)}(X) = \frac{d_D(n)}{d_D(m)} \Pi_{\text{sym},m}^{(D)}(X \otimes I_D^{\otimes(m-n)}) \Pi_{\text{sym},m}^{(D)}, \quad m \geq n. \quad (91)$$

If  $\rho$  has rank at most  $r$ , choose a purification  $\psi \in \mathbb{C}^d \otimes E$ ,  $E = \mathbb{C}^r$ . All purifications in this space differ by a unitary on  $E$ . Covariance of Werner’s cloner and invariance of partial trace therefore remove the Haar average in random purification exactly:

$$\sigma_{n,m}^{\text{PCT}}(\rho) = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(dr)}(|\psi\rangle\langle\psi|^{\otimes n}). \quad (92)$$

**Occupation and coherent-state limits.** Fix  $D \geq 2$ , a unit vector  $\psi \in \mathbb{C}^D$ , and an orthonormal basis  $e_1, \dots, e_{D-1}$  of  $K = \psi^\perp$ . For  $\mathbf{k} = (k_1, \dots, k_{D-1})$ , let  $|\mathbf{k}\rangle_m$  be the normalized symmetric vector with  $k_j$  particles in  $e_j$  and the remaining  $m - |\mathbf{k}|$  in  $\psi$ .

**Lemma A.1** (Werner occupation law). *The output  $\mathcal{W}_{n,m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n})$  is diagonal in this occupation basis, with eigenvalue*

$$\pi_{n,m}^{(D)}(\mathbf{k}) = \frac{\binom{m-|\mathbf{k}|}{n}}{\binom{m+D-1}{n+D-1}}, \quad |\mathbf{k}| \leq m - n. \quad (93)$$

If  $m_n/n \rightarrow \gamma > 1$ , its pullback  $\omega_{n,m_n}$  to the symmetric Fock space  $\Gamma_s(K)$  over  $K$ , with occupation basis  $|\mathbf{k}\rangle$ , satisfies

$$\left\| \omega_{n,m_n} - \tau_{(\gamma-1)/\gamma}^{\otimes(D-1)} \right\|_1 \rightarrow 0. \quad (94)$$

*Proof.* On an occupation vector with total transverse occupation  $t = |\mathbf{k}|$ , the expectation of  $\Pi_{\text{sym},m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n} \otimes I) \Pi_{\text{sym},m}^{(D)}$  is  $\binom{m-t}{n} / \binom{m}{n}$ . Multiplying by  $d_D(n)/d_D(m)$  gives (93). The stabilizer  $U(D-1)$  acts irreducibly on each  $\text{Sym}^t(K)$ , and its central phase eliminates cross- $t$  blocks, so these are all the eigenvalues. For each fixed  $\mathbf{k}$ ,

$$\pi_{n,m_n}^{(D)}(\mathbf{k}) \rightarrow \gamma^{-(D-1)} \left( \frac{\gamma-1}{\gamma} \right)^{|\mathbf{k}|}.$$

The limit is a normalized product of geometric laws. The discrete Scheffé lemma therefore gives  $\ell^1$  convergence, which is (94).  $\square$

Let  $R_M$  project onto Fock occupations of total number at most  $M$ , and let  $J_M|\mathbf{k}\rangle = |\mathbf{k}\rangle_M$  on this subspace. The trace-preserving occupation embedding is

$$\mathcal{J}_M(X) = J_M R_M X R_M J_M^* + \text{Tr}[(I - R_M)X] |\psi\rangle\langle\psi|^{\otimes M}. \quad (95)$$

For  $Z \in K$ , distinguish its coherent Fock vector  $|\text{coh}(Z)\rangle$  from the one-particle vector  $|Z\rangle$ , and put

$$|\psi_{Z,M}\rangle = \frac{|\psi\rangle + M^{-1/2}|Z\rangle}{\sqrt{1 + \|Z\|^2/M}}.$$

For every fixed  $Z$ ,

$$\left\| \mathcal{J}_M(|\text{coh}(Z)\rangle\langle\text{coh}(Z)|) - |\psi_{Z,M}\rangle\langle\psi_{Z,M}|^{\otimes M} \right\|_1 \longrightarrow 0. \quad (96)$$

Indeed, with  $t = \|Z\|^2$ , the product and coherent vectors have excitations only in the direction  $Z/\|Z\|$  (the case  $Z = 0$  is immediate). After pulling the product vector back by  $J_M^*$ , their number amplitudes are the square roots of the binomial and Poisson masses

$$B_M(j) = \binom{M}{j} \frac{(t/M)^j}{(1 + t/M)^M}, \quad P(j) = e^{-t} \frac{t^j}{j!}.$$

Pointwise convergence and the discrete Scheffé lemma give

$$\sum_{j \geq 0} (\sqrt{B_M(j)} - \sqrt{P(j)})^2 \leq \sum_{j \geq 0} |B_M(j) - P(j)| \longrightarrow 0.$$

Thus the pure states converge in trace norm; applying the contraction  $\mathcal{J}_M$  proves (96).

For  $s > 0$ , let  $\mu_s$  be the circular complex Gaussian measure on  $K$  with covariance  $sI_K$ , so each orthonormal coordinate has  $\mathbb{E}|Z_j|^2 = s$ . Gaussian integration in the number basis gives the positive coherent-state representation

$$\tau_{s/(1+s)}^{\otimes(D-1)} = \int_K |\text{coh}(Z)\rangle\langle\text{coh}(Z)| d\mu_s(Z), \quad d\mu_s(Z) = \frac{e^{-\|Z\|^2/s}}{(\pi s)^{D-1}} dZ. \quad (97)$$

Specifically, the integral is diagonal with eigenvalue  $s^{|\mathbf{k}|}/(1+s)^{|\mathbf{k}|+D-1}$ . Combining (92), (94), and (96) therefore gives, for any fixed purification  $\psi \in \mathbb{C}^d \otimes E$ ,

$$\left\| \sigma_{n,m_n}^{\text{PCT}}(\rho) - \int_K \rho_{Z,m_n}^{\otimes m_n} d\mu_{\gamma-1}(Z) \right\|_1 \longrightarrow 0, \quad \rho_{Z,M} = \text{Tr}_E |\psi_{Z,M}\rangle\langle\psi_{Z,M}|. \quad (98)$$

Here the finite Werner output equals  $\mathcal{J}_{m_n}(\omega_{n,m_n})$ . Contractivity controls its replacement by the thermal state, and dominated convergence controls the coherent-state integral: each trace-distance integrand is at most 2. No uniform estimate over unbounded  $Z$  is needed.

## A.2 Simple full-rank spectra

Fix  $p \in \Delta_d^\circ$ , so that  $\rho_p = \text{diag}(p_1, \dots, p_d)$  with  $p_1 > \dots > p_d > 0$ , and recall  $q_{ij} = p_j/p_i$ . Put

$$h_q(x) = \frac{\sqrt{(1-q)(1-x)}}{1 - \sqrt{qx}}, \quad (99)$$

and define

$$q_{ij}^{\text{PCT}} = \frac{\gamma - 1 + \gamma q_{ij}}{\gamma + (\gamma - 1)q_{ij}}. \quad (100)$$

The asymptotic PCT fidelity (75) can equivalently be written as

$$F_{\text{PCT}}(\gamma, p) = \left( \frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(d-1)/2} \prod_{i < j} h_{q_{ij}}(q_{ij}^{\text{PCT}}). \quad (101)$$

PCT here uses rank bound  $d$ , hence purification dimension  $d^2$ .



**Theorem A.2** (Exact full-rank fidelities and comparison). *For fixed  $d$ ,  $p$ , and  $\gamma > 1$  as above,*

$$\lim_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}\right) = F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p) = \lim_{n \rightarrow \infty} F\left(\mathcal{C}_n^{\text{univ}}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}\right), \quad (102)$$

where  $q_{ij} = p_j/p_i$ ,

$$F_{\text{PCT}}(\gamma, p) = \left(\frac{\sqrt{2\gamma-1}}{\gamma}\right)^{(d-1)/2} \prod_{i < j} \frac{\sqrt{\gamma + (\gamma-1)q_{ij}} + \sqrt{q_{ij}(\gamma-1 + \gamma q_{ij})}}{\gamma(1 + q_{ij})},$$

and  $F_{\text{univ}}$  and  $f_\gamma$  are as in (5) and (4).

*Proof.* The final equality is part (a) of Theorem 1.2. We prove the PCT limit by applying interior LAN to the product-state mixture (98), and then compare its factors.

*Step 1: The Gaussian output.* Choose

$$S = \text{diag}(\sqrt{p_1}, \dots, \sqrt{p_d}), \quad |\psi_p\rangle = \sum_i \sqrt{p_i} |ii\rangle, \quad K_p = \psi_p^\perp.$$

Identify the tangent vector  $|Z\rangle = \sum_{i,j} Z_{ij} |ij\rangle$  with its coefficient matrix. Partial trace has differential

$$D_p(Z) = ZS + SZ^*, \quad \rho_{Z,M} = \rho_p + M^{-1/2} D_p(Z) + O_Z(M^{-1}). \quad (103)$$

Its diagonal and orbital LAN coordinates are

$$h_i(Z) = 2\sqrt{p_i} \Re Z_{ii}, \quad \sum_i h_i(Z) = 0, \quad (104)$$

$$\beta_{ij}(Z) = \frac{D_p(Z)_{ji}}{\sqrt{p_i - p_j}} = \frac{\sqrt{p_i} Z_{ji} + \sqrt{p_j} \overline{Z_{ij}}}{\sqrt{p_i - p_j}}, \quad i < j. \quad (105)$$

To justify using LAN for  $\rho_{Z,M}$ , the chart  $(u, w) \mapsto e^{Y_p(w)} \text{diag}(p+u) e^{-Y_p(w)}$  from Section 3.1 has derivative  $(u, w) \mapsto \text{diag}(u) + [Y_p(w), \rho_p]$ , a real linear isomorphism onto the traceless Hermitian matrices. The inverse function theorem and (103) give exact local coordinates

$$\rho_{Z,M} = g_{M,z_M}^{(p)} \text{diag}(p + M^{-1/2} h_M) (g_{M,z_M}^{(p)})^*, \quad (h_M, z_M) = (h(Z), \beta(Z)) + O_Z(M^{-1/2}).$$

Let  $T_M^p, S_M^p$  be the forward and reverse interior LAN channels centered at  $\rho_p$ , in the normalization of (47), and set

$$\Theta(Z) = \mathbf{N}_{h(Z), \Sigma_p} \otimes \bigotimes_{i < j} D(\beta_{ij}(Z)) \tau_{q_{ij}} D(\beta_{ij}(Z))^*. \quad (106)$$

The bounded-parameter LAN estimates (48)–(49) and trace-norm continuity of Gaussian shifts imply, for each fixed  $Z$ ,

$$\left\| T_M^p(\rho_{Z,M}^{\otimes M}) - \Theta(Z) \right\|_1 \rightarrow 0, \quad \left\| \rho_{Z,M}^{\otimes M} - S_M^p(\Theta(Z)) \right\|_1 \rightarrow 0.$$

It remains to average  $\Theta(Z)$  under  $Z \sim \mu_s$ ,  $s = \gamma - 1$ . The orthogonal decomposition

$$K_p = \left\{ \sum_i z_i |ii\rangle : \sum_i \sqrt{p_i} z_i = 0 \right\} \oplus \bigoplus_{i < j} \text{span}\{|ij\rangle, |ji\rangle\}$$

makes the diagonal and pair variables independent. From (104)–(105),

$$\text{Cov}(h) = 2s\Sigma_p, \quad \mathbb{E}|\beta_{ij}|^2 = \frac{s(p_i + p_j)}{p_i - p_j}, \quad \mathbb{E}\beta_{ij}^2 = 0. \quad (107)$$

Here the diagonal Gaussian has covariance  $s(I - \sqrt{p}\sqrt{p}^\top)$ ; multiplying its real parts by  $2\sqrt{p_i}$  gives the first identity on the trace-zero hyperplane. Each  $\beta_{ij}$  is a circular complex Gaussian. Averaging the corresponding displaced thermal state therefore adds  $\mathbb{E}|\beta_{ij}|^2$  to its mean photon number and leaves it thermal: its Weyl characteristic function is multiplied by  $\exp(-\mathbb{E}|\beta_{ij}|^2|\xi|^2)$ . The output mean is

$$\bar{n}_{ij}^{\text{PCT}} = \frac{p_j + s(p_i + p_j)}{p_i - p_j} = \frac{(\gamma - 1)p_i + \gamma p_j}{p_i - p_j}, \quad (108)$$

whose geometric parameter is (100). Consequently,

$$\Theta_p^{\text{PCT}} := \int_{K_p} \Theta(Z) d\mu_{\gamma-1}(Z) = \mathbf{N}_{0, (2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}^{\text{PCT}}}. \quad (109)$$

Integrating both LAN approximations by dominated convergence and using (98) gives

$$\left\| T_{m_n}^p(\sigma_{n, m_n}^{\text{PCT}}(p)) - \Theta_p^{\text{PCT}} \right\|_1 \rightarrow 0, \quad (110)$$

$$\left\| \sigma_{n, m_n}^{\text{PCT}}(p) - S_{m_n}^p(\Theta_p^{\text{PCT}}) \right\|_1 \rightarrow 0. \quad (111)$$

The target has the same two-sided approximations with

$$\Theta_p = \Theta(0) = \mathbf{N}_{0, \Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}}. \quad (112)$$

*Step 2: Fidelity and strict comparison.* Monotonicity under  $T_{m_n}^p$  and  $S_{m_n}^p$ , together with trace-norm continuity, now gives the two-sided bound

$$F(\Theta_p^{\text{PCT}}, \Theta_p) - o(1) \leq F(\sigma_{n, m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) \leq F(\Theta_p^{\text{PCT}}, \Theta_p) + o(1).$$

The limiting fidelity factorizes, with

$$F(\mathbf{N}_{0, (2\gamma-1)\Sigma_p}, \mathbf{N}_{0, \Sigma_p}) = \left( \frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(d-1)/2} \quad (113)$$

and  $F(\tau_{q_{ij}^{\text{PCT}}}, \tau_{q_{ij}}) = h_{q_{ij}}(q_{ij}^{\text{PCT}})$ . This proves the exact value (101).

For one mode with  $q \in (0, 1)$ , the optimal amplifier has mean  $\bar{n}^{\text{opt}} = \gamma q / (1 - q) + \gamma - 1$ , hence

$$\bar{n}^{\text{opt}} > \frac{q}{1 - q}, \quad \bar{n}^{\text{PCT}} - \bar{n}^{\text{opt}} = \frac{(\gamma - 1)q}{1 - q} > 0.$$

Thus  $q < q^{\text{opt}} < q^{\text{PCT}}$ , where  $q^{\text{opt}} = (\gamma - 1 + q)/\gamma$  and  $f_\gamma(q) = h_q(q^{\text{opt}})$ . Since

$$\frac{d}{dx} \log h_q(x) = \frac{\sqrt{q} - \sqrt{x}}{2\sqrt{x}(1-x)(1-\sqrt{qx})} < 0, \quad x > q,$$

every PCT quantum factor is strictly smaller. PCT's classical factor is also smaller:  $c \mapsto 2\sqrt{c}/(1+c)$  decreases for  $c > 1$ , and  $2\gamma - 1 > \gamma > 1$ . Multiplying the factors proves  $F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p)$ .  $\square$

### A.3 Projector states

For projector states, we compare the fidelity attained by PCT itself with the optimum in [Theorem 1.3](#), rather than relying only on its published data-processing lower bound.

Fix  $1 \leq r < d$  and  $P \in \text{Gr}(r, d)$ , choose an isometry  $V_P : \mathbb{C}^r \rightarrow \mathbb{C}^d$  with  $V_P V_P^* = P$ , and put

$$|\Phi_P\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r V_P |i\rangle \otimes |i\rangle, \quad |\Phi_r\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r |ii\rangle.$$

Let  $\sigma_{n,m}^{\text{PCT}}(P)$  denote the system output of PCT on  $(P/r)^{\otimes n}$ , using a purifying register  $\mathbb{C}^r$ .

**Proposition A.3** (PCT is strictly suboptimal for non-pure projector states). *Define*

$$s_{n,m}^{(r,d)} = \frac{\mathbf{d}_{rd}(n)/\mathbf{d}_{rd}(m)}{\mathbf{d}_{r^2}(n)/\mathbf{d}_{r^2}(m)} = \prod_{j=r^2}^{rd-1} \frac{n+j}{m+j}. \quad (114)$$

*There is an exact finite-sample factorization*

$$F\left(\sigma_{n,m}^{\text{PCT}}(P), (P/r)^{\otimes m}\right) = \sqrt{s_{n,m}^{(r,d)}} F\left(\zeta_{n,m}^{(r)}, (I_r/r)^{\otimes m}\right), \quad (115)$$

where

$$\zeta_{n,m}^{(r)} = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(r^2)}(|\Phi_r\rangle\langle\Phi_r|^{\otimes n}). \quad (116)$$

If  $m_n/n \rightarrow \gamma > 1$ , then

$$\limsup_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n}\right) \leq \gamma^{-r(d-r)/2} \left(\frac{\sqrt{2\gamma-1}}{\gamma}\right)^{(r-1)/2}. \quad (117)$$

Consequently, for every  $1 < r < d$  and fixed  $\gamma > 1$ , PCT is strictly below the optimal projector-state value (9). For  $r = 1$ , the internal factor in (115) is one and PCT reduces to Werner's optimal pure-state cloner.

*Proof.* By (92), the PCT output is the reduced Werner output for the fixed purification  $\Phi_P$ . Let  $Q = P \otimes I_r$  on  $\mathbb{C}^d \otimes \mathbb{C}^r$ . Its rank is  $r^2$ , its range contains  $\Phi_P$ , and  $Q^{\otimes m}$  commutes with the symmetric projection. Compressing (91) to its range replaces the dimension  $rd$  by  $r^2$ , with normalization ratio  $s_{n,m}^{(r,d)}$ . Tracing out  $E^m$  therefore gives

$$P^{\otimes m} \sigma_{n,m}^{\text{PCT}}(P) P^{\otimes m} = s_{n,m}^{(r,d)} V_P^{\otimes m} \zeta_{n,m}^{(r)} (V_P^*)^{\otimes m}. \quad (118)$$

The target is supported on  $P^{\otimes m}$ . Hence fidelity is unchanged by compressing the first argument to that support, and its homogeneity proves (115).

It remains to bound the internal fidelity. Assume  $r > 1$ , put

$$\mathbf{H}_r = \left\{ h \in \mathbb{R}^r : \sum_{i=1}^r h_i = 0 \right\}, \quad u_r = \frac{1}{r} \mathbf{1}_r, \quad \Sigma_r = (\text{diag}(u_r) - u_r u_r^\top) \Big|_{\mathbf{H}_r},$$

and measure every system output in the computational basis. If  $C_m = (C_{m,1}, \dots, C_{m,r})$  is the resulting count vector, set

$$Z_m = \frac{C_m - m u_r}{\sqrt{m}} \in \mathbf{H}_r.$$

Under the target  $(I_r/r)^{\otimes m}$ , the multinomial central limit theorem gives

$$Z_m \implies \mathbf{N}_{0, \Sigma_r}. \quad (119)$$

We next show that under  $\zeta_{n, m_n}^{(r)}$ ,

$$Z_{m_n} \implies \mathbf{N}_{0, (2\gamma-1)\Sigma_r}. \quad (120)$$

Apply (98) to the internal state, with condensate  $\Phi_r$ , dimension  $r^2$ , and  $Z \sim \mu_{\gamma-1}$  on  $\Phi_r^\perp$ . Identify  $Z$  with a traceless complex  $r \times r$  matrix. Conditional on  $Z$ , measuring the system part of the product of the normalized vector  $\Phi_r + m^{-1/2}Z$  gives multinomial probabilities

$$\begin{aligned} p_i^{(m)}(Z) &= \frac{r^{-1} + 2\Re Z_{ii}/\sqrt{rm} + m^{-1} \sum_j |Z_{ij}|^2}{1 + m^{-1} \|Z\|^2} \\ &= \frac{1}{r} + \frac{h_i(Z)}{\sqrt{m}} + O_Z(m^{-1}), \quad h_i(Z) = \frac{2\Re Z_{ii}}{\sqrt{r}}. \end{aligned} \quad (121)$$

The multinomial central limit theorem therefore gives the conditional limit  $\mathbf{N}_{h(Z), \Sigma_r}$ . Since  $\text{Tr } Z = 0$ , the mean  $h(Z)$  belongs to  $\mathbf{H}_r$ ; under  $\mu_{\gamma-1}$  it is centered Gaussian, with

$$\text{Cov}(h(Z))_{ij} = \frac{2(\gamma-1)}{r} \left( \delta_{ij} - \frac{1}{r} \right) = 2(\gamma-1)(\Sigma_r)_{ij}.$$

Integrating the conditional characteristic functions, justified by dominated convergence, gives covariance  $\Sigma_r + 2(\gamma-1)\Sigma_r = (2\gamma-1)\Sigma_r$ . The trace-norm approximation (98) and contractivity of measurement transfer this limit to  $\zeta_{n, m_n}^{(r)}$ , proving (120).<sup>5</sup>

Let  $L_n \Rightarrow L$  and  $T_n \Rightarrow T$  be the count laws and their Gaussian limits. For any finite partition  $\mathcal{P}$  whose boundaries have zero Gaussian measure, coarse-graining and weak convergence give

$$\limsup_n F(L_n, T_n) \leq \sum_{A \in \mathcal{P}} \sqrt{L(A)T(A)}.$$

Along a refining sequence generating the Borel sets, the right-hand side decreases to  $F(L, T)$ , proving the required upper semicontinuity. Combining this with fidelity monotonicity under the count measurement and (119)–(120) therefore gives

$$\limsup_{n \rightarrow \infty} F(\zeta_{n, m_n}^{(r)}, (I_r/r)^{\otimes m_n}) \leq \left( \frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(r-1)/2}. \quad (122)$$

Finally, (114) gives

$$s_{n, m_n}^{(r, d)} \longrightarrow \gamma^{-r(d-r)}.$$

Combining this with (115) and (122) proves (117). Its right-hand side is strictly smaller than  $\gamma^{-r(d-r)/2}$  when  $1 < r < d$ , because  $2\gamma-1 < \gamma^2$ . The  $r=1$  assertion follows directly from (116).  $\square$

**Proposition A.4** (PCT's leading projector small-error coefficient). *Fix  $1 < r < d$  and  $P \in \text{Gr}(r, d)$ , put  $a = r(d-r)$ , and, for each fixed  $0 < \delta < 1/(2r)$ , take  $m_n/n \rightarrow 1 + \delta$ . Define*

$$\mathcal{E}_n(\delta) = 1 - F(\sigma_{n, m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n})^2, \quad C_r(\delta) = (1 - 4\delta^2)^{-(r^2-1)/2} - 1.$$

<sup>5</sup>An alternative proof uses the count fields. For  $h \in \mathbf{H}_r$ , put  $H_h = \text{diag}(h) \otimes I_r$ . Under the occupation embedding, the rescaled observable  $m^{-1/2} \sum_\ell H_h^{(\ell)}$ , extended by zero outside the occupation cutoff, converges on the analytic finite-particle core to  $b^*(H_h \Phi_r) + b(H_h \Phi_r)$ . The strong-resolvent core criterion [RS80, Theorem VIII.25] then gives the same thermal characteristic-function limit.

Then

$$\begin{aligned} 1 - (1 + \delta)^{-a} &\leq \liminf_n \mathcal{E}_n(\delta) \leq \limsup_n \mathcal{E}_n(\delta) \\ &\leq 1 - (1 + \delta)^{-a} + (1 + \delta)^{-a} C_r(\delta). \end{aligned} \quad (123)$$

Thus both envelopes equal  $a\delta + O_{r,d}(\delta^2)$  as  $\delta \downarrow 0$ . In this iterated limit, PCT has the same leading small-error coefficient as the optimal projector cloner, despite its strictly smaller fidelity at each fixed gain.

*Proof.* It suffices to bound the internal squared infidelity in (115). Specialize the mixture (98) to the condensate  $\Phi_r$ , and write

$$\tilde{\zeta}_M = \int_{\Phi_r^\perp} \rho_{Z,M}^{\otimes M} d\mu_\delta(Z), \quad \rho_{Z,M} = \frac{I_r}{r} + \frac{A_Z}{\sqrt{M}} + O_Z(M^{-1}), \quad A_Z = \frac{Z + Z^*}{\sqrt{r}}.$$

Here  $Z$  is a traceless complex  $r \times r$  matrix, and  $\|\zeta_{n,m_n}^{(r)} - \tilde{\zeta}_{m_n}\|_1 \rightarrow 0$ . For independent  $Z, W \sim \mu_\delta$ , the scaled purity satisfies

$$\begin{aligned} r^M \text{Tr} \tilde{\zeta}_M^2 &= \mathbb{E} \left[ (r \text{Tr}(\rho_{Z,M} \rho_{W,M}))^M \right], \\ (r \text{Tr}(\rho_{Z,M} \rho_{W,M}))^M &\longrightarrow e^{r \text{Tr}(A_Z A_W)}. \end{aligned}$$

To justify exchanging limit and expectation, contraction under partial trace and the pure-state trace-distance formula give

$$\left\| \rho_{Z,M} - \frac{I_r}{r} \right\|_2 \leq \| |\psi_{Z,M}\rangle\langle\psi_{Z,M}| - |\Phi_r\rangle\langle\Phi_r| \|_1 = 2\sqrt{\frac{\|Z\|^2}{M + \|Z\|^2}}.$$

Since both perturbations have trace zero, Cauchy–Schwarz and  $1 + x \leq e^x$  yield

$$0 \leq (r \text{Tr}(\rho_{Z,M} \rho_{W,M}))^M \leq e^{4r\|Z\|\|W\|} \leq e^{2r(\|Z\|^2 + \|W\|^2)}.$$

The last expression is integrable when  $\delta < 1/(2r)$ . In a Hilbert–Schmidt orthonormal basis of traceless Hermitian matrices, the real parts  $x_j, y_j$  of the coordinates of  $Z, W$  are independent  $\mathbf{N}_{0,\delta/2}$  variables and  $r \text{Tr}(A_Z A_W) = 4 \sum_{j=1}^{r^2-1} x_j y_j$ . Dominated convergence and the Gaussian cross-moment therefore give

$$\lim_{M \rightarrow \infty} r^M \text{Tr} \tilde{\zeta}_M^2 = \prod_{j=1}^{r^2-1} \mathbb{E} e^{4x_j y_j} = (1 - 4\delta^2)^{-(r^2-1)/2} = 1 + C_r(\delta). \quad (124)$$

For any state  $\sigma$  on  $(\mathbb{C}^r)^{\otimes M}$ ,

$$1 - F\left(\sigma, (I_r/r)^{\otimes M}\right)^2 \leq r^M \text{Tr} \sigma^2 - 1.$$

Indeed, apply  $(\sqrt{x} - 1)^2 \leq (x - 1)^2$  to the relative eigenvalues  $x = r^M \lambda_j(\sigma)$ , average with weights  $r^{-M}$ , and use  $1 - F^2 \leq 2(1 - F)$ . Applying this inequality to  $\tilde{\zeta}_M$  and then using fidelity continuity to transfer the bound to the PCT output gives

$$\limsup_n \left[ 1 - F\left(\zeta_{n,m_n}^{(r)}, (I_r/r)^{\otimes m_n}\right)^2 \right] \leq C_r(\delta). \quad (125)$$

Finally, the exact support factorization and  $s_{n,m_n}^{(r,d)} \rightarrow (1 + \delta)^{-a}$  imply (123). Since  $C_r(\delta) = 2(r^2 - 1)\delta^2 + O_r(\delta^4)$ , the asserted first-order coefficient follows. For any fixed  $0 < \eta < 1$  and sufficiently small  $\epsilon > 0$ , both envelopes lie below  $\epsilon$  at  $\delta = (1 - \eta)\epsilon/a$  and above it at  $\delta = (1 + \eta)\epsilon/a$ . Letting  $\eta \downarrow 0$  gives the leading sample coefficient  $a$  in Section 5, without assuming convergence of the fidelity at fixed  $\delta$ .  $\square$

For completeness, the subleading terms in the two explicitly invertible threshold formulas of [Section 5](#) follow from the elementary expansion, valid for any  $a > 0$ ,

$$\frac{1}{(1-\epsilon)^{-1/a}-1} = \frac{a}{\epsilon} - \frac{a+1}{2} + O(\epsilon). \quad (126)$$

Taking  $a = r(d-r)$  gives the optimal limiting projector threshold [\(84\)](#); taking  $a = rd-1$  gives the published PCT sufficient threshold [\(85\)](#). [Proposition A.4](#) identifies PCT's actual leading coefficient, but not its subleading threshold terms or exact internal fidelity.

## B Exact-weight PBW calculation of the sector fidelity

This appendix proves [Proposition 2.3](#) by comparing the target sector  $\rho_{p,\nu}$  with the Cartan-sector output  $\mathcal{C}_{\mu,\nu}(\rho_{p,\mu})$  on finitely many weight spaces near their highest weights, then controlling the remaining target mass. [Lemma B.1](#) shows that normalized lowering-operator vectors become asymptotically orthonormal on these spaces, while [Lemma B.2](#) gives the uniform limit  $p^\lambda/s_\lambda(p) \rightarrow \prod_{i < j} (1 - p_j/p_i)$ . The Cartan inclusion  $V_{\mu,\nu-\mu}$  then splits each lowering operation between  $\mathcal{H}_\mu$  and  $\mathcal{H}_{\nu-\mu}$  with asymptotically binomial weights, producing the thermal limits  $\tau_p$  and  $\tau_p^{(\gamma)}$  and hence the fidelity  $F_{\text{orb}}(\gamma, p)$ .

Because the calculations involve only weights of height bounded in terms of  $L$ ,  $O_L(\cdot)$  may depend on  $L, d, c, C$  in  $cN \leq \Delta_\alpha^\lambda \leq CN$  for  $\alpha \in \mathbb{R}_+$ , but is uniform in  $N$  and all dominant  $\lambda$  satisfying these inequalities.

### B.1 PBW vectors on complete weight spaces

Identify the positive roots of  $\mathbb{U}(d)$  with

$$\mathbb{R}_+ = \{e_i - e_j : 1 \leq i < j \leq d\}.$$

For  $\alpha = e_i - e_j$ , set

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij}, \quad H_\alpha = [E_\alpha, F_\alpha] = E_{ii} - E_{jj}, \quad \Delta_\alpha^\lambda = \lambda_i - \lambda_j.$$

Fix a total order on  $\mathbb{R}_+$ . For  $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbb{R}_+}$ , write

$$F^{\mathbf{r}} = \prod_{\alpha \in \mathbb{R}_+}^< F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_{\alpha} r_\alpha.$$

For a nonnegative simple-root combination  $\eta$ , let  $\text{ht}(\eta)$  be its simple-root height and put

$$\mathbb{P}(\eta) = \left\{ \mathbf{r} : \sum_{\alpha \in \mathbb{R}_+} r_\alpha \alpha = \eta \right\}.$$

The set  $\mathbb{P}(\eta)$  is finite, and the PBW theorem implies that  $\{F^{\mathbf{r}} v_\lambda : \mathbf{r} \in \mathbb{P}(\eta)\}$  spans  $\mathcal{H}_\lambda[\lambda - \eta]$  [\[FH91\]](#). Define

$$\mathcal{K}_{\lambda,L} = \bigoplus_{\text{ht}(\eta) \leq L} \mathcal{H}_\lambda[\lambda - \eta]. \quad (127)$$

Let  $P_{\lambda,L}$  denote the orthogonal projection onto  $\mathcal{K}_{\lambda,L}$ .

**Lemma B.1** (PBW Gram matrices at fixed height). *Let  $\lambda = \lambda^{(N)}$  range over dominant weights satisfying*

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in \mathbf{R}_+)$$

*for fixed  $0 < c < C < \infty$ . Define*

$$\tilde{e}_\mathbf{r}^\lambda = \prod_{\alpha \in \mathbf{R}_+}^< \frac{F_\alpha^{r_\alpha}}{\sqrt{r_\alpha!} (\Delta_\alpha^\lambda)^{r_\alpha/2}} v_\lambda. \quad (128)$$

*For every fixed  $L$ , uniformly over  $\text{ht}(\eta) \leq L$  and  $\mathbf{r}, \mathbf{s} \in \mathbf{P}(\eta)$ ,*

$$\langle \tilde{e}_\mathbf{r}^\lambda, \tilde{e}_\mathbf{s}^\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} + O_L(N^{-1/2}). \quad (129)$$

*Consequently, for all sufficiently large  $N$ , the full Gram matrix  $G_{\lambda, \eta}$  on every such exact-weight block is invertible. If*

$$(e_\mathbf{r}^\lambda)_{\mathbf{r} \in \mathbf{P}(\eta)} = (\tilde{e}_\mathbf{r}^\lambda)_{\mathbf{r} \in \mathbf{P}(\eta)} G_{\lambda, \eta}^{-1/2}, \quad (130)$$

*then these vectors form an orthonormal basis of  $\mathcal{H}_\lambda[\lambda - \eta]$ , and*

$$e_\mathbf{r}^\lambda = \tilde{e}_\mathbf{r}^\lambda + O_L(N^{-1/2}) \quad (131)$$

*uniformly for  $\text{ht}(\eta) \leq L$  and  $\mathbf{r} \in \mathbf{P}(\eta)$ .*

*Proof.* For  $\mathbf{r} \in \mathbf{P}(\eta)$ , every positive root has positive simple-root height, so  $|\mathbf{r}| \leq \text{ht}(\eta)$ . Put

$$\hat{F}_\alpha = \frac{F_\alpha}{\sqrt{\Delta_\alpha^\lambda}}, \quad \hat{E}_\alpha = \frac{E_\alpha}{\sqrt{\Delta_\alpha^\lambda}}.$$

Each unnormalized Gram entry  $\langle v_\lambda, (\hat{F}^\mathbf{r})^* \hat{F}^\mathbf{s} v_\lambda \rangle$  is therefore a word of length at most  $2L$ . Commuting root operators cannot increase the word length, and each root has height at most  $d - 1$ . Hence every intermediate vector lies in

$$\mathcal{K}_{\lambda, R_L}, \quad R_L := 2L(d - 1).$$

The normalized root operators are uniformly bounded on each fixed cutoff. Indeed, put  $h = \text{ht}(\alpha)$ , set  $\mathcal{K}_{\lambda, R} = \{0\}$  for  $R < 0$ , and write

$$M_R = \|\hat{F}_\alpha|_{\mathcal{K}_{\lambda, R}}\|^2.$$

Since the cutoffs contain complete weight spaces,  $\hat{E}_\alpha : \mathcal{K}_{\lambda, R} \rightarrow \mathcal{K}_{\lambda, R-h}$  is the adjoint of  $\hat{F}_\alpha : \mathcal{K}_{\lambda, R-h} \rightarrow \mathcal{K}_{\lambda, R}$ , and hence its squared norm is  $M_{R-h}$ . On  $\mathcal{K}_{\lambda, R}$ ,  $[\hat{E}_\alpha, \hat{F}_\alpha] = H_\alpha / \Delta_\alpha^\lambda = I + O_R(N^{-1})$ . Applying this identity to unit vectors gives

$$M_R \leq M_{R-h} + 1 + O_R(N^{-1}) \leq \lfloor R/h \rfloor + 1 + O_R(N^{-1}) = O_R(1),$$

where the second inequality follows by induction from  $M_R = 0$  for  $R < 0$ . This also bounds  $\hat{E}_\alpha$ , without using the Gram estimate. All operator norms below are taken after restriction to  $\mathcal{K}_{\lambda, R_L}$ , with the codomain norm inherited from  $\mathcal{H}_\lambda$ .

The matrix-unit commutator formula now gives

$$[\hat{E}_\alpha, \hat{F}_\beta] = \delta_{\alpha\beta} I + O_L(N^{-1/2}). \quad (132)$$

For  $\alpha = \beta$ , the left side is  $H_\alpha/\Delta_\alpha^\lambda = I + O_L(N^{-1})$  on the cutoff. For  $\alpha \neq \beta$ , the unnormalized commutator is zero or, up to sign, a root operator  $E_\gamma$  or  $F_\gamma$ . In the nonzero case, the preceding bound gives  $\|[\widehat{E}_\alpha, \widehat{F}_\beta]\| = O_L(\sqrt{\Delta_\gamma^\lambda}/\sqrt{\Delta_\alpha^\lambda \Delta_\beta^\lambda}) = O_L(\sqrt{N}/N) = O_L(N^{-1/2})$ .

Commute every  $\widehat{E}_\alpha$  to the right in  $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$ . The scalar commutators in (132) contribute only when  $\mathbf{r} = \mathbf{s}$ , in which case they give  $\prod_\alpha r_\alpha!$ . Every other term vanishes or contains an  $O_L(N^{-1/2})$  remainder. Only  $O_L(1)$  terms occur, and all remaining operator factors have norm  $O_L(1)$ ; consequently,

$$\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} \prod_\alpha r_\alpha! + O_L(N^{-1/2}).$$

Dividing by the factorials in (128) proves (129).

The number and sizes of the exact-weight blocks are bounded in terms of  $L$ , so the entrywise estimate gives  $G_{\lambda, \eta} = I + O_L(N^{-1/2})$  in operator norm. Thus  $G_{\lambda, \eta}$  is invertible for large  $N$ , and the PBW spanning vectors form a basis. Finally, functional calculus gives  $G_{\lambda, \eta}^{-1/2} = I + O_L(N^{-1/2})$ ; hence (130) is an orthonormal basis and satisfies (131).  $\square$

## B.2 Character normalization and fixed-height matrices

For  $a = (a_1, \dots, a_d)$ , write  $p^a = \prod_i p_i^{a_i}$ .

**Lemma B.2** (Uniform character normalization). *Let  $K \in \Delta_d^\circ$ , let  $\delta_N \rightarrow 0$ , and suppose  $p \in K$ ,  $\lambda \in Y_N$ , and  $\|\lambda/N - p\|_\infty \leq \delta_N$ . Then, uniformly over this family,*

$$\frac{p^\lambda}{s_\lambda(p)} = \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right) + o(1). \quad (133)$$

*Proof.* Write the numerator in the Weyl character formula as

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma, \quad A_\sigma = \prod_i p_i^{\lambda_{\sigma(i)} + d - \sigma(i)}.$$

For every nonidentity permutation, strict rearrangement and compactness of  $K$  give, uniformly for all sufficiently large  $N$ ,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} = \sum_i (p_i - p_{\sigma(i)}) \log p_i + O_K(\delta_N + N^{-1}) \geq c_K > 0.$$

Thus the numerator is  $A_{\text{id}}(1 + O_K(e^{-c_K N}))$ . Cancelling  $\prod_i p_i^{d-i}$  against the Weyl denominator gives

$$s_\lambda(p) = p^\lambda \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right)^{-1} (1 + O_K(e^{-c_K N})).$$

This proves (133).  $\square$

For the remainder of the appendix, index the number basis of  $\mathcal{F}_{\text{orb}}$  by  $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbf{R}_+}$ , and write it as  $\{|\mathbf{r}\rangle\}$ . Define the finite-rank projection

$$P_L^{\mathcal{F}} = \sum_{\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L} |\mathbf{r}\rangle \langle \mathbf{r}|. \quad (134)$$



For every fixed  $L$  and all sufficiently large  $N$ , each highest weight  $\lambda$  satisfying the gap bounds in [Lemma B.1](#) has the orthonormal basis  $e_{\mathbf{r}}^\lambda$  of  $\mathcal{K}_{\lambda,L}$  constructed there. Define

$$J_{\lambda,L} : \mathcal{K}_{\lambda,L} \longrightarrow P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}, \quad J_{\lambda,L} e_{\mathbf{r}}^\lambda = |\mathbf{r}\rangle. \quad (135)$$

By [Lemma B.1](#), the two spaces have bases indexed by the same  $\mathbf{r}$ , so  $J_{\lambda,L}$  is unitary onto  $P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}$ . It is defined only on  $\mathcal{K}_{\lambda,L}$ , and it depends on  $\lambda$  and  $L$ .

*Proof of Proposition 2.3.* Fix  $K$  and  $\delta_n$  as in the proposition, write  $m = m_n$ , and put  $\omega = \nu - \mu$ . Since  $m/n = \gamma_n \rightarrow \gamma > 1$ , we have  $m/n - 1 \geq (\gamma - 1)/2$  for all sufficiently large  $n$ . Uniformly over the allowed parameters,

$$\left\| \frac{\omega}{m-n} - p \right\|_\infty \leq \frac{m/n + 1}{m/n - 1} \delta_n = o(1).$$

Every coordinate and every adjacent gap is uniformly positive on  $K$ . Hence  $\omega \vdash m - n$  for all sufficiently large  $n$ . For  $\alpha = e_i - e_j$ , set

$$s_{\alpha,n} := \frac{\Delta_\alpha^\mu}{\Delta_\alpha^\nu} = \frac{n}{m} \frac{\mu_i/n - \mu_j/n}{\nu_i/m - \nu_j/m} = \frac{1}{\gamma} + o(1), \quad 1 - s_{\alpha,n} = \frac{\Delta_\alpha^\omega}{\Delta_\alpha^\nu}. \quad (136)$$

Consequently, there are constants  $0 < c_K < C_K < \infty$ , depending only on  $K$  and  $\gamma$ , such that, for all large  $n$ ,

$$c_K n \leq \Delta_\alpha^\mu, \Delta_\alpha^\nu, \Delta_\alpha^\omega \leq C_K n \quad (\alpha \in R_+). \quad (137)$$

Let  $V = V_{\mu,\omega} : \mathcal{H}_\nu \rightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega$ , and write  $F_\alpha^{(\lambda)} = d\pi_\lambda(F_\alpha)$  for the action of the root operator on  $\mathcal{H}_\lambda$ . Intertwining and the highest-weight normalization give

$$VF_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})V.$$

Expanding each  $(F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})^{r_\alpha}$  binomially gives the exact identity

$$V\tilde{e}_{\mathbf{r}}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \tilde{e}_{\mathbf{a}}^\mu \otimes \tilde{e}_{\mathbf{r}-\mathbf{a}}^\omega, \quad (138)$$

where

$$\beta_{\mathbf{r},\mathbf{a}}^{(n)} = \prod_{\alpha \in R_+} \sqrt{\binom{r_\alpha}{a_\alpha} s_{\alpha,n}^{a_\alpha} (1 - s_{\alpha,n})^{r_\alpha - a_\alpha}}. \quad (139)$$

If  $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$ , then (138) has  $O_L(1)$  terms, all of height at most  $L$ , with  $0 \leq \beta_{\mathbf{r},\mathbf{a}}^{(n)} \leq 1$ . Since  $m/n \rightarrow \gamma > 1$ , the sizes  $n, m, m - n$  are uniformly comparable, so (137) and [Lemma B.1](#) give  $e_{\mathbf{b}}^\lambda = \tilde{e}_{\mathbf{b}}^\lambda + O_L(n^{-1/2})$  for  $\lambda \in \{\mu, \nu, \omega\}$  and all relevant  $\mathbf{b}$ . Substituting these relations into (138) and using  $\|Vx\| = \|x\|$  gives

$$Ve_{\mathbf{r}}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} e_{\mathbf{a}}^\mu \otimes e_{\mathbf{r}-\mathbf{a}}^\omega + O_L(n^{-1/2}) \quad (140)$$

uniformly in norm for  $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$ .

Put

$$\sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \quad c_\lambda(p) = \frac{p^\lambda}{s_\lambda(p)}, \quad q_\alpha = \frac{p_j}{p_i} \quad (\alpha = e_i - e_j).$$

For  $\lambda \in \{\mu, \nu\}$ , the weight of  $e_{\mathbf{b}}^\lambda$  is  $\lambda - \sum_\alpha b_\alpha \alpha$ , and hence

$$\rho_{p,\lambda} e_{\mathbf{b}}^\lambda = c_\lambda(p) \prod_\alpha q_\alpha^{b_\alpha} e_{\mathbf{b}}^\lambda.$$

The sector channel gives the local formula

$$\sigma_{\mu,\nu} = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V^* (\rho_{p,\mu} \otimes I_{\mathcal{H}_\omega}) V.$$

All asymptotic estimates below are uniform over the allowed family. The character estimate, [Lemma B.2](#), gives

$$c_\lambda(p) = \prod_\alpha (1 - q_\alpha) + o(1) \quad (\lambda = \mu, \nu),$$

while the Weyl dimension formula, (137), and (136) give

$$\frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} = \prod_{\alpha=e_i-e_j} \frac{\Delta_\alpha^\mu + j - i}{\Delta_\alpha^\nu + j - i} = \prod_{\alpha \in \mathbb{R}_+} (s_{\alpha,n} + O(n^{-1})) = \gamma^{-r_d} + o(1).$$

Substitute (140) into the sector formula. In the resulting double sum, the weight action and orthonormality force  $\mathbf{a} = \mathbf{b}$  in the input factor and then  $\mathbf{r} = \mathbf{s}$  in the auxiliary factor. The binomial theorem and  $\|\rho_{p,\mu}\| \leq 1$  therefore give, for fixed  $L$ ,

$$\langle e_{\mathbf{r}}^\nu, \sigma_{\mu,\nu} e_{\mathbf{s}}^\nu \rangle = \delta_{\mathbf{r},\mathbf{s}} \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} c_\mu(p) \prod_{\alpha \in \mathbb{R}_+} (1 - s_{\alpha,n} + s_{\alpha,n} q_\alpha)^{r_\alpha} + O_L(n^{-1/2}), \quad (141)$$

$$\langle e_{\mathbf{r}}^\nu, \rho_{p,\nu} e_{\mathbf{s}}^\nu \rangle = \delta_{\mathbf{r},\mathbf{s}} c_\nu(p) \prod_{\alpha \in \mathbb{R}_+} q_\alpha^{r_\alpha}. \quad (142)$$

Recall the thermal notation in (31) and (30):

$$q_\alpha^{(\gamma)} = 1 - \frac{1 - q_\alpha}{\gamma}, \quad \tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle \langle k|, \quad \tau_p = \bigotimes_{\alpha \in \mathbb{R}_+} \tau_{q_\alpha}, \quad \tau_p^{(\gamma)} = \bigotimes_{\alpha \in \mathbb{R}_+} \tau_{q_\alpha^{(\gamma)}}.$$

Since  $s_{\alpha,n} = \gamma^{-1} + o(1)$ , the limiting entries in (141) and (142) are those of  $\tau_p^{(\gamma)}$  and  $\tau_p$ , respectively. The cutoff space has fixed finite dimension, so this uniform entrywise convergence gives

$$\left\| J_{\nu,L} P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0. \quad (143)$$

$$\left\| J_{\nu,L} P_{\nu,L} \rho_{p,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0. \quad (144)$$

To obtain the fidelity assertions, let  $(a_\mu)_\mu$  be a probability distribution supported on labels  $\mu \in Y_n$  with  $\|\mu/n - p\|_\infty \leq \delta_n$ . All such pairs  $(\mu, \nu)$  are compatible for large  $n$ , by the first part of the proof. Put

$$\bar{\sigma}_{p,\nu} = \sum_\mu a_\mu \sigma_{\mu,\nu}, \quad \sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}).$$

For fixed  $L$ , let  $\varepsilon_{n,L} \rightarrow 0$  bound both trace-norm errors in (143) and (144), uniformly over the allowed parameters. Crucially, the output frame  $J_{\nu,L} P_{\nu,L}$  is independent of  $\mu$ . Convexity of the trace norm therefore gives the same error bound for  $\bar{\sigma}_{p,\nu}$ , uniformly over all mixture weights.

Next we control the target mass above height  $L$ . Since  $K$  is compact in  $\Delta_d^\circ$ , there is  $b_K < 1$  such that  $q_\alpha \leq b_K$  for every  $p \in K$  and every positive root  $\alpha$ . Since  $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq (d-1)r_d \max_\alpha r_\alpha$  and each occupation under  $\tau_p$  has geometric tail  $\Pr(r_\alpha \geq k) = q_\alpha^k$ , a union bound gives

$$t_L := \sup_{p \in K} \text{Tr}[(I - P_L^\mathcal{F})\tau_p] \leq r_d b_K^{\lfloor L/((d-1)r_d) \rfloor + 1} \longrightarrow 0 \quad (L \rightarrow \infty). \quad (145)$$

Taking traces in (144) gives

$$\text{Tr}[(I - P_{\nu,L})\rho_{p,\nu}] \leq t_L + \varepsilon_{n,L}. \quad (146)$$

Every  $\sigma_{\mu,\nu}$ , and hence  $\bar{\sigma}_{p,\nu}$ , commutes with the diagonal torus, by torus invariance of  $\rho_{p,\mu}$  and covariance of  $\mathcal{C}_{\mu,\nu}$ ; so does  $\rho_{p,\nu}$ . Consequently,  $P_{\nu,L}$  reduces both states, and the direct-sum identity splits their fidelity into the sum of the cutoff-block and complementary-block fidelities. Schatten Hölder gives  $F(A, C) \leq \sqrt{\text{Tr } A \text{ Tr } C}$ , so by (146) and  $\text{Tr}[(I - P_{\nu,L})\bar{\sigma}_{p,\nu}] \leq 1$  the complementary fidelity is at most  $\sqrt{t_L + \varepsilon_{n,L}}$ . The same decomposition for  $F(\tau_p^{(\gamma)}, \tau_p)$  with respect to  $P_L^\mathcal{F}$  bounds its complementary part by  $\sqrt{t_L}$ . Finally, fidelity is invariant under the isometry  $J_{\nu,L}$ , so the continuity estimate (36) and the two  $\varepsilon_{n,L}$ -bounds above show that the two cutoff fidelities differ by at most  $2\sqrt{\varepsilon_{n,L}}$ . Altogether,

$$\left| F(\bar{\sigma}_{p,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \leq 2\sqrt{\varepsilon_{n,L}} + \sqrt{t_L + \varepsilon_{n,L}} + \sqrt{t_L}.$$

First let  $n \rightarrow \infty$  at fixed  $L$ , then let  $L \rightarrow \infty$ ; the convergence is uniform over  $p \in K$ , over the admissible labels, and over the weights  $(a_\mu)$ . Finally,

$$F(\tau_p^{(\gamma)}, \tau_p) = \prod_{i < j} F(\tau_{q_{ij}}^{(\gamma)}, \tau_{q_{ij}}) = F_{\text{orb}}(\gamma, p),$$

where multiplicativity of fidelity follows from multiplicativity of the trace norm under tensor products, and the last equality uses (39) and (42). This proves the mixture assertion of Proposition 2.3; taking a point mass gives the individual-sector assertion.  $\square$

## C Young local limit and randomized rounding

This appendix proves Proposition 2.4. We embed Young-label laws as densities on lattice cells and prove their uniform Gaussian limit. Randomized rounding then becomes exactly dilation followed by cell averaging, which yields the classical factor  $F_{\text{cl}}(\gamma, d)$ . The fallback rule changes only the vanishing mass of atypical input labels.

Define the half-open fundamental cell

$$\mathbf{C} = \left\{ \sum_{i=1}^{d-1} t_i(e_i - e_d) : -\frac{1}{2} \leq t_i < \frac{1}{2} \right\} \subset \mathbf{H}.$$

Volumes are taken with the induced  $(d-1)$ -dimensional Lebesgue measure. The rounding rule (25) is equivalent, outside a null set of cell boundaries, to choosing the unique  $\tilde{\nu} \in \mathbf{L}_{m_n}$  for which

$$\gamma_n(\mu + U) \in \tilde{\nu} + \mathbf{C}, \quad U = \sum_{i=1}^{d-1} U_i(e_i - e_d).$$

The first  $d-1$  coordinates give the nearest-integer rule, and the last is fixed by the coordinate sum. Since  $U$  is uniform on  $\mathbf{C}$ ,

$$D_n(\tilde{\nu} \mid \mu) = \frac{|\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})|}{\gamma_n^{d-1} |\mathbf{C}|}. \quad (147)$$

All label laws below are regarded as functions on  $\mathbf{L}_N$ : if a law  $R$  is originally defined on  $\mathbf{Y}_N$ , set  $R(\lambda) = 0$  for every  $\lambda \in \mathbf{L}_N \setminus \mathbf{Y}_N$ .

For  $\lambda \in \mathbf{L}_N$ , define the recentered and rescaled lattice cell

$$\mathbf{C}_{N,p}(\lambda) = \frac{\lambda - Np + \mathbf{C}}{\sqrt{N}}.$$

For any  $R : \mathbf{L}_N \rightarrow \mathbb{R}$  satisfying  $\sum_{\lambda \in \mathbf{L}_N} |R(\lambda)| < \infty$ , define

$$(\mathcal{E}_{N,p}R)(x) = \frac{N^{(d-1)/2}}{|\mathbf{C}|} R(\lambda) \quad (x \in \mathbf{C}_{N,p}(\lambda)).$$

The cells  $\mathbf{C}_{N,p}(\lambda)$ ,  $\lambda \in \mathbf{L}_N$ , tile  $\mathbf{H}$ , and their boundaries have zero induced Lebesgue measure. The map  $\mathcal{E}_{N,p}$  preserves  $\ell^1$ -distance,  $\|\mathcal{E}_{N,p}R - \mathcal{E}_{N,p}S\|_1 = \|R - S\|_{\ell^1}$ , and, for nonnegative  $R, S$ , Hellinger affinity,  $\int_{\mathbf{H}} \sqrt{(\mathcal{E}_{N,p}R)(\mathcal{E}_{N,p}S)} \, dx = \sum_{\lambda \in \mathbf{L}_N} \sqrt{R(\lambda)S(\lambda)}$ . Indeed, each  $\mathbf{C}_{N,p}(\lambda)$  has volume  $|\mathbf{C}|N^{-(d-1)/2}$ , which cancels the prefactor in both cellwise integrals.

**Lemma C.1** (Uniform Young local limit). *For every compact  $K \Subset \Delta_d^\circ$ ,*

$$\sup_{p \in K} \|\mathcal{E}_{N,p}P_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (148)$$

*Proof.* Write  $f_{N,p} = \mathcal{E}_{N,p}P_{N,p}$ , and fix a ball  $B_R = \{x \in \mathbf{H} : \|x\| \leq R\}$ . If  $\mathbf{C}_{N,p}(\lambda)$  meets  $B_R$ , then  $x_\lambda = (\lambda - Np)/\sqrt{N}$  stays bounded uniformly in  $N, p$ . Since the smallest eigenvalue and the adjacent gaps of  $p \in K$  are bounded below, every such  $\lambda$  is a Young diagram for large  $N$ . The Frobenius dimension formula and Lemma B.2 give, uniformly on these cells,

$$P_{N,p}(\lambda) = \frac{N! \prod_i p_i^{\lambda_i}}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)} \prod_i \prod_{a=1}^{d-i} \frac{Np_i}{\lambda_i + a} (1 + o(1)).$$

Each factor in the two last products is  $1 + O_{K,R}(N^{-1/2})$ . Stirling's formula and a Taylor expansion of the first factor therefore yield

$$P_{N,p}(\lambda) = \frac{\exp\left[-\frac{1}{2} \sum_i x_{\lambda,i}^2 / p_i\right]}{(2\pi N)^{(d-1)/2} \sqrt{\prod_i p_i}} (1 + o(1)).$$

The normalization agrees with the Gaussian density on  $\mathbf{H}$ :

$$|\mathbf{C}| = \sqrt{d}, \quad \det_{\mathbf{H}} \Sigma_p = d \prod_i p_i, \quad \langle x, \Sigma_p^{-1} x \rangle = \sum_i \frac{x_i^2}{p_i} \quad (x \in \mathbf{H}).$$

Indeed, the Gram matrix of  $(e_i - e_d)_{i < d}$  has determinant  $d$ . The ambient covariance  $\text{diag}(p) - pp^\top$  has kernel  $\text{span}\{(1, \dots, 1)\}$  and principal cofactors  $\prod_i p_i$ , giving its restricted determinant. Its inverse on  $\mathbf{H}$  sends  $x$  to  $\text{diag}(p)^{-1}x$  up to a multiple of  $(1, \dots, 1)$ , giving the quadratic identity. The cell diameter is  $O(N^{-1/2})$ , so uniform continuity of the Gaussian densities on bounded sets gives

$$\sup_{p \in K} \sup_{x \in B_R} |f_{N,p}(x) - \varphi_{0,\Sigma_p}(x)| \longrightarrow 0.$$

Both densities have integral one. Consequently,

$$\|f_{N,p} - \varphi_{0,\Sigma_p}\|_1 \leq 2 \int_{B_R} |f_{N,p} - \varphi_{0,\Sigma_p}| \, dx + 2 \int_{B_R^c} \varphi_{0,\Sigma_p}(x) \, dx.$$

Let  $N \rightarrow \infty$ , then  $R \rightarrow \infty$ . The Gaussian tails vanish uniformly on compact  $K$ , proving (148).  $\square$

**Lemma C.2** (Continuous limit of randomized dilation). *Let  $\tilde{Q}_{m_n,p} = P_{n,p}D_n$  denote the  $\tilde{\nu}$ -marginal of the joint law obtained by composing the kernel  $D_n$  with  $P_{n,p}$ , before the replacement in (26). For every compact  $K \subseteq \Delta_d^\circ$ ,*

$$\sup_{p \in K} \|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma\Sigma_p}\|_1 \longrightarrow 0. \quad (149)$$

$$\sup_{p \in K} |F(\tilde{Q}_{m_n,p}, P_{m_n,p}) - F_{\text{cl}}(\gamma, d)| \longrightarrow 0. \quad (150)$$

*Proof.* For  $a > 0$  and  $f \in L^1(\mathbf{H})$ , set

$$(\text{dil}_a f)(y) = a^{-(d-1)/2} f(y/\sqrt{a}),$$

and define the target-cell averaging operator explicitly by

$$(\text{Av}_{m_n,p} f)(x) := \frac{m_n^{(d-1)/2}}{|\mathbf{C}|} \int_{\mathbf{C}_{m_n,p}(\nu)} f(y) \, dy, \quad x \in \mathbf{C}_{m_n,p}(\nu).$$

If  $X = (\mu - np + U)/\sqrt{n}$ , where  $\mu \sim P_{n,p}$  and  $U$  is uniform on  $\mathbf{C}$ , then  $X$  has density  $\mathcal{E}_{n,p}P_{n,p}$ . Since  $m_n = \gamma_n n$ , the event  $\sqrt{\gamma_n}X \in \mathbf{C}_{m_n,p}(\nu)$  is exactly the rounding event  $\gamma_n(\mu + U) \in \nu + \mathbf{C}$ . Thus the dilated density assigns mass  $\tilde{Q}_{m_n,p}(\nu)$  to that target cell, and

$$\text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p}P_{n,p}) = \mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p}. \quad (151)$$

Cell averaging is an  $L^1$ -contraction and dilation an  $L^1$ -isometry. Since  $\text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} = \varphi_{0,\gamma_n \Sigma_p}$ , (151) yields

$$\|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma\Sigma_p}\|_1 \leq \|\mathcal{E}_{n,p}P_{n,p} - \varphi_{0,\Sigma_p}\|_1 + \|\text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma\Sigma_p}\|_1 + \|\varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma\Sigma_p}\|_1.$$

The first term tends uniformly to zero by Lemma C.1, and the last does so because  $\gamma_n \rightarrow \gamma$ . For the middle term, the cellwise  $L^1$  Poincaré inequality gives, for smooth  $f$  with integrable gradient,

$$\|\text{Av}_{m_n,p} f - f\|_1 \leq C_d m_n^{-1/2} \|\nabla f\|_1.$$

Gaussian gradients have uniformly bounded  $L^1$ -norm on the compact covariance family  $\gamma_n \Sigma_p$ , so this term is  $O_K(m_n^{-1/2})$ . This proves (149).

Applying Lemma C.1 at sample size  $m_n$  gives

$$\mathcal{E}_{m_n,p} P_{m_n,p} \longrightarrow \varphi_{0,\Sigma_p}$$

uniformly in  $L^1$ . The embedding preserves Hellinger affinity, and (36) also holds for nonnegative  $L^1$  densities. Therefore

$$F(\tilde{Q}_{m_n,p}, P_{m_n,p}) \longrightarrow F(\mathbf{N}_{0,\gamma\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = F_{\text{cl}}(\gamma, d)$$

uniformly on  $K$ . □

*Proof of Proposition 2.4.* Write  $e = \tilde{\nu} - \gamma_n \mu$ . The nearest-integer rule gives  $|e_i| \leq (\gamma_n + 1)/2$  for  $i < d$ , and  $e_d = -\sum_{i < d} e_i$ . Thus  $\|e\|_\infty \leq c_{\text{rnd}}$ , where

$$c_{\text{rnd}} = \frac{d-1}{2} \left( 1 + \sup_n \gamma_n \right) < \infty.$$

Compactness of  $K$  gives  $c_0 > 0$  such that, for all large  $n$ , every  $p \in K$  and  $\mu \in \mathbf{T}_n(p)$  satisfy

$$\mu_d \geq c_0 n, \quad \mu_i - \mu_{i+1} \geq c_0 n \quad (i < d).$$

For  $\omega = \tilde{\nu} - \mu = (\gamma_n - 1)\mu + e$ , these bounds imply

$$\omega_d \geq (\gamma_n - 1)c_0n - c_{\text{rnd}}, \quad \omega_i - \omega_{i+1} \geq (\gamma_n - 1)c_0n - 2c_{\text{rnd}} \quad (i < d).$$

Since  $\gamma_n \rightarrow \gamma > 1$ , both are positive for large  $n$ . Thus  $\omega \vdash m_n - n$  and  $\tilde{\nu} = \mu + \omega \in \mathbf{Y}_{m_n}$ . No rounded label is changed for typical  $\mu$ , proving (32).

Coupling the raw and final labels by the fallback rule therefore gives

$$\|Q_{m_n,p} - \tilde{Q}_{m_n,p}\|_{\ell^1} \leq 2P_{n,p}(\mathbf{T}_n(p)^c) = o(1)$$

uniformly on  $K$ , by Lemma 2.1. Combining (150) with (36) proves (33).  $\square$

## D Unified Gaussian flat-prior converse

This appendix proves the two converse inequalities in Theorem 3.1. The weighted-fidelity bound (Lemma D.1) reduces fidelity to an oscillator moment; compact-observable limits (Lemma D.2) and the least-noise inequality (Proposition D.3) bound that moment. Proposition D.4 combines these ingredients with a direct Gaussian translation estimate for the classical factor, giving  $F_{\text{orb}}(\gamma, p)$  in the orbital model and  $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p)$  in the full model.

The classical parameter space has dimension  $k = 0$  for the orbital problem and  $k = d - 1$  for the full problem. All trace-class-valued integrals below are Bochner integrals in trace norm. For classical–quantum maps, complete positivity means positivity of  $\text{id}_{M_a} \otimes \Lambda$  for every  $a \geq 1$ , with the matrix factor acting on the quantum trace-class fibers; the classical algebra is abelian.

### D.1 Weighted fidelity and compact limits

**Lemma D.1** (Weighted fidelity bound). *Let  $R, T$  be positive trace-class operators, and let  $W$  be bounded, positive, and injective. Define*

$$\text{Tr}(TW^{-1}) := \lim_{\varepsilon \downarrow 0} \text{Tr}(T(W + \varepsilon I)^{-1}) \in [0, \infty].$$

*If this quantity is finite, then*

$$F(R, T)^2 \leq \text{Tr}(RW) \text{Tr}(TW^{-1}). \quad (152)$$

*Let  $\mathbf{E}$  be a finite-dimensional Euclidean space, let  $R \in L^1(\mathbf{E}; \mathcal{T}_1(\mathcal{F}_{\text{orb}}))$  be positive, let  $g$  be a probability density on  $\mathbf{E}$ , and let  $\chi > 0$  be measurable. For positive  $R, S$  in this  $L^1$ -space, set  $F(R, S) = \int_{\mathbf{E}} F(R(y), S(y)) \, dy$ , extending the hybrid-state convention in Section 3. The notation  $gT$  means the density  $y \mapsto g(y)T$ . Whenever the right-hand side below is finite,*

$$F(R, gT)^2 \leq \left( \int_{\mathbf{E}} \chi(y) \text{Tr}[R(y)W] \, dy \right) \left( \int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right) \text{Tr}(TW^{-1}). \quad (153)$$

*Proof.* For  $W_\varepsilon = W + \varepsilon I$ , the Schatten Hölder inequality gives

$$F(R, T) = \|(R^{1/2}W_\varepsilon^{-1/2})(W_\varepsilon^{-1/2}T^{1/2})\|_1 \leq \sqrt{\text{Tr}(RW_\varepsilon)} \sqrt{\text{Tr}(TW_\varepsilon^{-1})}.$$

Squaring and letting  $\varepsilon \downarrow 0$ , using  $\text{Tr}(RW_\varepsilon) = \text{Tr}(RW) + \varepsilon \text{Tr} R$  and monotone convergence for the second trace, proves (152). For the hybrid state,

$$F(R, gT) = \int_{\mathbf{E}} \sqrt{g(y)} F(R(y), T) \, dy.$$

Apply the first inequality pointwise and then Cauchy–Schwarz to the factors  $\sqrt{\chi(y) \text{Tr}[R(y)W]}$  and  $\sqrt{g(y)/\chi(y)}$ . This gives (153).  $\square$

**Lemma D.2** (Compact-observable limits of completely positive maps). *Let  $\mathcal{H}_0, \mathcal{H}_1$  be separable Hilbert spaces and let  $\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$  be completely positive maps with  $\sup_j \|\Gamma_j\|_{1 \rightarrow 1} < \infty$ . There are a subsequence and a completely positive map  $\Gamma : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$  such that*

$$\mathrm{Tr}[\Gamma_j(X)K] \longrightarrow \mathrm{Tr}[\Gamma(X)K] \quad (154)$$

for every  $X \in \mathcal{T}_1(\mathcal{H}_0)$  and every compact  $K \in \mathcal{K}(\mathcal{H}_1)$ . If  $\mathrm{Tr} \Gamma_j(X) \leq c \mathrm{Tr} X$  for every  $X \geq 0$ , then  $\mathrm{Tr} \Gamma(X) \leq c \mathrm{Tr} X$ .

Suppose in addition that  $U_g \in \mathcal{U}(\mathcal{H}_0)$  and  $V_g \in \mathcal{U}(\mathcal{H}_1)$  are families of unitaries indexed by a set  $G$ , and that, for every fixed  $g \in G$  and  $X \in \mathcal{T}_1(\mathcal{H}_0)$ ,

$$\|\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*\|_1 \longrightarrow 0.$$

Then

$$\Gamma(U_g X U_g^*) = V_g \Gamma(X) V_g^* \quad (155)$$

for every  $g \in G$  and  $X \in \mathcal{T}_1(\mathcal{H}_0)$ .

*Proof.* Let  $M = \sup_j \|\Gamma_j\|_{1 \rightarrow 1}$ , and let  $\mathcal{D} \subset \mathcal{K}(\mathcal{H}_1)$  be a countable norm-dense set. The predual  $\mathcal{T}_1(\mathcal{H}_0)$  is separable, so a diagonal weak-\* compactness argument gives a subsequence for which  $\Gamma_j^*(K)$  converges weak-\* in  $\mathcal{B}(\mathcal{H}_0)$  for every  $K \in \mathcal{D}$ . The bound  $\|\Gamma_j^*(K)\| \leq M\|K\|$  extends both the limit and the convergence to every compact  $K$ , producing a bounded complex-linear map  $\Gamma^* : \mathcal{K}(\mathcal{H}_1) \rightarrow \mathcal{B}(\mathcal{H}_0)$ .

For every positive block matrix  $A = [K_{ab}]$  of compact operators,  $[\Gamma_j^*(K_{ab})] \geq 0$ ; its entrywise weak-\* limit is  $[\Gamma^*(K_{ab})] \geq 0$ . Thus  $\Gamma^*$  is completely positive.

Since  $\mathcal{K}(\mathcal{H}_1)^* = \mathcal{T}_1(\mathcal{H}_1)$ , define  $\Gamma(X)$  by

$$\mathrm{Tr}[\Gamma(X)K] = \mathrm{Tr}[X\Gamma^*(K)] \quad (K \in \mathcal{K}(\mathcal{H}_1)).$$

Then  $\Gamma$  is completely positive and (154) holds. If  $P_N \uparrow I$  are finite-rank projections and  $X \geq 0$ , then

$$\mathrm{Tr} \Gamma(X) = \sup_N \lim_j \mathrm{Tr}[\Gamma_j(X)P_N] \leq c \mathrm{Tr} X.$$

Finally, for fixed  $g, X$  and compact  $K$ , the operator  $V_g^* K V_g$  is compact. Passing to the limit gives

$$\mathrm{Tr}[\Gamma(U_g X U_g^*)K] = \mathrm{Tr}[V_g \Gamma(X) V_g^* K].$$

Compact operators separate trace-class operators, so the limiting map is covariant.  $\square$

## D.2 The least-noise oscillator moment

Fix  $\gamma > 1$ . For  $s \geq 1$  and  $q_1, \dots, q_s \in (0, 1)$ , put

$$q_\ell^{(\gamma)} = 1 - \frac{1 - q_\ell}{\gamma}, \quad \tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell^{(\gamma)}}.$$

The multimode idler representation expresses the output number law as a mixture of product negative-binomial laws. A direct coupling then gives the least-noise moment, without phase averaging or a mode-by-mode argument.

**Fact D.I** (Covariant-amplifier representation). *Let  $G_1, \dots, G_s > 1$ , and write  $\sqrt{\mathbf{G}}z = (\sqrt{G_\ell}z_\ell)_{\ell=1}^s$ . Every  $s$ -mode channel  $\Lambda$  satisfying*

$$\Lambda(D(z)XD(z)^*) = D(\sqrt{\mathbf{G}}z)\Lambda(X)D(\sqrt{\mathbf{G}}z)^*$$

*admits an idler realization*

$$a_{\ell,\text{out}} = \sqrt{G_\ell} a_{\ell,\text{in}} + \sqrt{G_\ell - 1} b_\ell^*, \quad \ell = 1, \dots, s. \quad (156)$$

*The dilation unitary is a product of two-mode squeezers. The joint idler state is uniquely determined by the channel and need not be Gaussian, number diagonal, or a product state.*

*Proof.* Covariance implies that  $\Lambda^*(D(w))D(\sqrt{\mathbf{G}}w)^*$  commutes with every displacement. Irreducibility of the Weyl system therefore gives

$$\Lambda^*(D(w)) = f(w)D(\sqrt{\mathbf{G}}w)$$

for a scalar function  $f$ . In real phase-space coordinates the associated defect symplectic form is  $\bigoplus_{\ell=1}^s (1 - G_\ell) \Omega_2$ , where  $\Omega_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ . It is nondegenerate, so the quantum Bochner criterion [Hol82] identifies  $f$ , after modewise rescaling and phase conjugation, with the characteristic function of a unique joint idler state. In the exact dilation theorem [LSW18, Lemma 7], take the system matrix  $\bigoplus_{\ell} \sqrt{G_\ell} I_2$  and the noise matrix  $\bigoplus_{\ell} \sqrt{G_\ell - 1} \text{diag}(1, -1)$ . These are precisely the system and idler blocks of the product of two-mode squeezers in (156). The one-mode gain-2 case is also derived directly in [GM06, Appendix A].  $\square$

**Proposition D.3** (Least-noise moment). *Let  $\Gamma$  be a completely positive trace-nonincreasing map on the  $s$ -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

*If*

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^s w_\ell(\hat{N}_\ell),$$

*where every  $w_\ell$  is bounded, nonnegative, and nonincreasing on  $\mathbb{Z}_{\geq 0}$ , then*

$$\text{Tr}[\Gamma(\tau)W(\hat{\mathbf{N}})] \leq \text{Tr}[\Gamma(\tau)] \text{Tr}[\tau^{(\gamma)}W(\hat{\mathbf{N}})]. \quad (157)$$

*Proof.* Displacement covariance makes  $\Gamma^*(I)$  commute with every Weyl displacement. Irreducibility therefore gives  $\Gamma^*(I) = cI$ , with  $c = \text{Tr} \Gamma(\tau) \in [0, 1]$ . The case  $c = 0$  is immediate; otherwise  $\Lambda = c^{-1}\Gamma$  is a channel with the same covariance.

The quantum-limited amplifier  $\mathcal{A}_G$  sends vacuum to  $\tau_{1-1/G}$ . Thus

$$\tau = \left( \bigotimes_{\ell=1}^s \mathcal{A}_{1/(1-q_\ell)} \right) (|\mathbf{0}\rangle\langle\mathbf{0}|). \quad (158)$$

The composite  $\mathcal{B} = \Lambda \circ \bigotimes_{\ell} \mathcal{A}_{1/(1-q_\ell)}$  sends joint vacuum to  $\Lambda(\tau)$  and has modewise gains

$$G_\ell = \frac{\gamma}{1 - q_\ell}, \quad \lambda_\ell = 1 - G_\ell^{-1} = q_\ell^{(\gamma)}.$$

Apply **Fact D.I** to  $\mathcal{B}$ , and let  $\sigma$  be its joint idler state.



For a single two-mode squeezer, normal-ordered disentangling [BR97] gives, with  $\lambda = 1 - G^{-1}$ ,

$$U_G|0, l\rangle = (1 - \lambda)^{(l+1)/2} \sum_{k \geq 0} \sqrt{\binom{k+l}{l}} \lambda^{k/2} |k, k+l\rangle. \quad (159)$$

In the product expansion, the  $\mathbf{k}$ -diagonal signal element survives the idler trace only when  $\mathbf{k} + \mathbf{l} = \mathbf{k} + \mathbf{l}'$ , hence  $\mathbf{l} = \mathbf{l}'$ . Only the joint diagonal  $a_{\mathbf{l}} = \langle \mathbf{l} | \sigma | \mathbf{l} \rangle$  contributes to the output number probabilities:

$$\langle \mathbf{k} | \Lambda(\tau) | \mathbf{k} \rangle = \sum_{\mathbf{l} \in \mathbb{Z}_{\geq 0}^s} a_{\mathbf{l}} \prod_{\ell=1}^s \binom{k_{\ell} + l_{\ell}}{l_{\ell}} (1 - \lambda_{\ell})^{l_{\ell}+1} \lambda_{\ell}^{k_{\ell}}. \quad (160)$$

Thus, conditional on  $\mathbf{l}$ , the coordinates are independent  $\text{NB}(l_{\ell} + 1, \lambda_{\ell})$  variables. The weights  $a_{\mathbf{l}}$  may be correlated; no product or phase-invariance assumption on the idler is needed.

Sample  $\mathbf{L}$  with law  $a_{\mathbf{l}}$ , independently of geometric variables  $Z_{\ell,j}$ , all independent, with  $\Pr(Z_{\ell,j} = k) = (1 - \lambda_{\ell}) \lambda_{\ell}^k$  for  $j, k \geq 0$ . The coupling

$$N_{\ell} = \sum_{j=0}^{L_{\ell}} Z_{\ell,j} \geq Z_{\ell,0} \quad (\ell = 1, \dots, s)$$

has the output law (160), while  $(Z_{\ell,0})_{\ell}$  has the number law of  $\tau^{(\gamma)}$ . Since each  $w_{\ell}$  is nonnegative and nonincreasing,

$$\text{Tr}[\Lambda(\tau)W] = \mathbb{E} \prod_{\ell} w_{\ell}(N_{\ell}) \leq \mathbb{E} \prod_{\ell} w_{\ell}(Z_{\ell,0}) = \text{Tr}[\tau^{(\gamma)}W].$$

Multiplying by  $c = \text{Tr} \Gamma(\tau)$  proves (157). The one-mode gain-2 stochastic order is also established in [GM06, Lemma 2.3].  $\square$

Specialize to  $s = r_d$ ,  $q_{\ell} = q_{ij}$ ,  $\tau = \tau_p$ , and  $\tau^{(\gamma)} = \tau_p^{(\gamma)}$ . For  $\mathbf{k} = (k_{ij})_{i < j} \in \mathbb{Z}_{\geq 0}^{r_d}$ , let  $t_{\mathbf{k}}$  and  $t_{\mathbf{k}}^{(\gamma)}$  be the corresponding number-basis eigenvalues, and define

$$W_{\gamma} = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (161)$$

Equivalently,

$$W_{\gamma} = \prod_{i < j} w_{ij}(\hat{N}_{ij}), \quad w_{ij}(n) = \sqrt{\frac{1 - q_{ij}}{1 - q_{ij}^{(\gamma)}}} \left( \frac{q_{ij}}{q_{ij}^{(\gamma)}} \right)^{n/2}.$$

Here  $\hat{N}_{ij}$  is the number operator of the mode indexed by  $i < j$ , and  $|\mathbf{k}| := \sum_{i < j} k_{ij}$  is the total occupation number. Since  $q_{ij} < q_{ij}^{(\gamma)}$ , each  $w_{ij}(n)$  is positive, bounded, and strictly decreasing in  $n$ , and the eigenvalues of  $W_{\gamma}$  tend to zero as  $|\mathbf{k}| \rightarrow \infty$ . Thus  $W_{\gamma}$  is bounded, injective, and compact. Its definition gives

$$\text{Tr}(\tau_p^{(\gamma)} W_{\gamma}) = \text{Tr}(\tau_p W_{\gamma}^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = F_{\text{orb}}(\gamma, p), \quad (162)$$

where the inverse is understood as in Lemma D.1.

### D.3 One flat-prior converse for both Gaussian models

**Proposition D.4** (Flat-prior converse inequality). *Let  $k \in \{0, d-1\}$ . For  $k=0$ , let  $\mathbf{E} = \{0\}$ , omit the classical component, and interpret  $h$ -integration as evaluation at 0. For  $k=d-1$ , let  $\mathbf{E} = \mathbf{H}$  with its induced Euclidean measure. Let*

$$\mathbf{E}_L = \{h \in \mathbf{E} : \|h\|_\infty \leq L\}, \quad \Xi_L = \mathbf{E}_L \times \mathbf{Z}_L,$$

with  $|\mathbf{E}_L| = 1$  when  $k=0$ . Define

$$R_{0,z}^{(0)} = \Phi_z^{(p)}, \quad S_{0,z}^{(0)} = \Psi_z^{(p,\gamma)},$$

$$R_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{in}}, \quad S_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{tar}}, \quad c_k = \left( \frac{2\sqrt{\gamma}}{1+\gamma} \right)^{k/2}.$$

For  $k=0$ , the supremum below is over quantum channels on  $\mathcal{F}_{\text{orb}}$ ; for  $k=d-1$ , it is over classical-quantum channels. Then

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_{h,z}^{(k)}), S_{h,z}^{(k)}) \, dh \, dz \leq c_k F_{\text{orb}}(\gamma, p). \quad (163)$$

*Proof.* For  $k=0$ , set  $g_1 = g_\gamma = \chi = 1$  on the one-point space. For  $k=d-1$ , let  $g_1$  and  $g_\gamma$  be the centered Gaussian densities with covariances  $\Sigma_p$  and  $\gamma\Sigma_p$ , and set

$$\chi(y) = \sqrt{\frac{g_1(y)}{g_\gamma(y)}} = \gamma^{k/4} \exp \left[ -\frac{\gamma-1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle \right].$$

This function is bounded and strictly positive. In both cases,

$$\int_{\mathbf{E}} \frac{g_1(y)}{\chi(y)} \, dy = \int_{\mathbf{E}} \sqrt{g_1(y)g_\gamma(y)} \, dy = c_k. \quad (164)$$

For  $k=d-1$ , completing the square also gives

$$H(r) := \int_{\mathbf{E}} g_1(h) \chi(r + \sqrt{\gamma}h) \, dh = c_k \exp \left[ -\frac{\gamma-1}{2\gamma(\gamma+1)} \langle r, \Sigma_p^{-1} r \rangle \right] \leq c_k. \quad (165)$$

For  $\xi = (h, z)$ , define

$$(\mathcal{U}_\xi^{\text{in}} R)(x) = D(z) R(x-h) D(z)^*, \quad (\mathcal{U}_\xi^{\text{out}} R)(y) = D(\sqrt{\gamma}z) R(y - \sqrt{\gamma}h) D(\sqrt{\gamma}z)^*.$$

Then  $R_\xi^{(k)} = \mathcal{U}_\xi^{\text{in}}(g_1 \tau_p)$  and  $S_\xi^{(k)} = \mathcal{U}_\xi^{\text{out}}(g_1 \tau_p)$ . For a channel  $\Lambda$ , define its average pointwise by

$$\bar{\Lambda}_L(R) = \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\eta}^{\text{out}}(\Lambda(\mathcal{U}_\eta^{\text{in}} R)) \, d\eta.$$

For each fixed input  $R$ , the integrand is trace-norm continuous and bounded in norm by  $\|R\|_1$ , so the Bochner integral is well-defined. Invariance of  $F$  under the isometries  $\mathcal{U}_\xi^{\text{out}}$  and joint concavity, applied to the Bochner average, give

$$F_L(\Lambda) = \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_\xi^{(k)}), S_\xi^{(k)}) \, d\xi \leq F(\bar{\Lambda}_L(g_1 \tau_p), g_1 \tau_p). \quad (166)$$

Let  $F_* = \limsup_{L \rightarrow \infty} \sup_{\Lambda} F_L(\Lambda)$ . Choose  $L_j \rightarrow \infty$  with  $\sup_{\Lambda} F_{L_j}(\Lambda) \rightarrow F_*$ , and channels  $\tilde{\Lambda}_j$  satisfying  $F_{L_j}(\tilde{\Lambda}_j) \geq \sup_{\Lambda} F_{L_j}(\Lambda) - j^{-1}$ . Set

$$\Lambda_j = \overline{\tilde{\Lambda}_{j_{L_j}}}, \quad R_j = \Lambda_j(g_1 \tau_p).$$

Then

$$F_{L_j}(\tilde{\Lambda}_j) \rightarrow F_*, \quad F_{L_j}(\tilde{\Lambda}_j) \leq F(R_j, g_1 \tau_p). \quad (167)$$

Write

$$A_{j,u} = \mathcal{U}_{-u}^{\text{out}} \circ \tilde{\Lambda}_j \circ \mathcal{U}_u^{\text{in}}.$$

The group law and the change of variables  $u = \eta + \xi$  give

$$\begin{aligned} \Lambda_j(\mathcal{U}_{\xi}^{\text{in}} R) &= \mathcal{U}_{\xi}^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j} + \xi} A_{j,u}(R) du, \\ \mathcal{U}_{\xi}^{\text{out}} \Lambda_j(R) &= \mathcal{U}_{\xi}^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,u}(R) du. \end{aligned}$$

The integrals cancel on the intersection of the two averaging sets. Since  $\mathcal{U}_{\xi}^{\text{out}}$  is a trace-norm isometry and each  $A_{j,u}$  is a channel, hence a trace-norm contraction, this gives

$$\|\Lambda_j(\mathcal{U}_{\xi}^{\text{in}} R) - \mathcal{U}_{\xi}^{\text{out}} \Lambda_j(R)\|_1 \leq b_j(\xi) \|R\|_1, \quad b_j(\xi) := \frac{|(\Xi_{L_j} + \xi) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|}. \quad (168)$$

Here  $0 \leq b_j \leq 2$ , and  $b_j(\xi) = O_{\xi}(L_j^{-1}) \rightarrow 0$  for fixed  $\xi$ , since  $\Xi_1$  is a bounded polytope and  $\Xi_L = L\Xi_1$ .

Define

$$\Gamma_j^{\chi}(X) = \int_{\mathbb{E}} \chi(y) [\Lambda_j(g_1 X)](y) dy.$$

These maps are completely positive and satisfy  $\|\Gamma_j^{\chi}\|_{1 \rightarrow 1} \leq \|\chi\|_{\infty}$ . We first show the sharper trace bound

$$\text{Tr } \Gamma_j^{\chi}(X) \leq (c_k + e_j) \text{Tr } X \quad (X \geq 0), \quad e_j := \|\chi\|_{\infty} \int_{\mathbb{E}} g_1(h) b_j(h, 0) dh \rightarrow 0. \quad (169)$$

For  $k = 0$ , this is trace preservation, with  $c_0 = 1$  and  $e_j = 0$ . For  $k = d - 1$ , take a nonnegative probability approximate identity  $u_{\varepsilon}$  on  $\mathbb{E}$  and put  $B_{j,\varepsilon} = \Lambda_j(u_{\varepsilon} X)$ , whose total trace is  $\text{Tr } X$ . Expressing  $g_1 * u_{\varepsilon}$  as an average of translates of  $u_{\varepsilon}$ , (168) gives

$$\left\| \Lambda_j((g_1 * u_{\varepsilon})X) - \int_{\mathbb{E}} g_1(h) \mathcal{U}_{(h,0)}^{\text{out}} B_{j,\varepsilon} dh \right\|_1 \leq \text{Tr } X \int_{\mathbb{E}} g_1(h) b_j(h, 0) dh.$$

Pairing with  $\chi$ , changing variables, and using (165) therefore yields

$$\int_{\mathbb{E}} \chi(y) \text{Tr}[\Lambda_j((g_1 * u_{\varepsilon})X)(y)] dy \leq \int_{\mathbb{E}} H(r) \text{Tr } B_{j,\varepsilon}(r) dr + e_j \text{Tr } X \leq (c_k + e_j) \text{Tr } X.$$

Letting  $\varepsilon \downarrow 0$  proves (169), since  $g_1 * u_{\varepsilon} \rightarrow g_1$  in  $L^1$ . Dominated convergence gives  $e_j \rightarrow 0$ .

Taking  $\xi = (0, z)$  in (168), multiplying by  $\chi$ , and integrating gives

$$\|\Gamma_j^{\chi}(D(z)XD(z)^*) - D(\sqrt{\gamma}z)\Gamma_j^{\chi}(X)D(\sqrt{\gamma}z)^*\|_1 \leq \|\chi\|_{\infty} b_j(0, z) \|X\|_1 \rightarrow 0.$$

Hence, after passing to a subsequence, [Lemma D.2](#) gives a completely positive, displacement-covariant limit  $\Gamma^\chi$  such that

$$\mathrm{Tr}[\Gamma_j^\chi(X)K] \longrightarrow \mathrm{Tr}[\Gamma^\chi(X)K] \quad (170)$$

for trace-class  $X$  and compact  $K$ . For  $X \geq 0$ , testing against finite-rank projections and using [\(169\)](#) gives

$$\mathrm{Tr} \Gamma^\chi(X) \leq c_k \mathrm{Tr} X.$$

In particular, the limiting map is trace nonincreasing. With

$$M_j = \int_{\mathbb{E}} \chi(y) \mathrm{Tr}[R_j(y)W_\gamma] \, dy = \mathrm{Tr}[\Gamma_j^\chi(\tau_p)W_\gamma],$$

compactness of  $W_\gamma$  gives

$$M_j \longrightarrow \mathrm{Tr}[\Gamma^\chi(\tau_p)W_\gamma]. \quad (171)$$

Moreover, [Lemma D.1](#), [\(164\)](#), and [\(162\)](#) give

$$F(R_j, g_1 \tau_p)^2 \leq M_j c_k F_{\mathrm{orb}}(\gamma, p). \quad (172)$$

By [Proposition D.3](#),

$$\lim_j M_j = \mathrm{Tr}[\Gamma^\chi(\tau_p)W_\gamma] \leq \mathrm{Tr} \Gamma^\chi(\tau_p) F_{\mathrm{orb}}(\gamma, p) \leq c_k F_{\mathrm{orb}}(\gamma, p).$$

Combining [\(167\)](#) with [\(172\)](#) yields  $F_* \leq c_k F_{\mathrm{orb}}(\gamma, p)$ , proving both cases.  $\square$

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